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Article

Noncommutative Nest Hardy Spaces and Martingales

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Let $(\mathcal{M},\tau)$ be a finite von Neumann algebra equipped with a normalized faithful trace and let $\mathbb{A}$ be a nest of order type $\mathbb{N}$. The nest algebra is defined by $H^r_\infty(\mathbb{A})=\{x\in\mathcal{M}: ex=exe, e\in\mathbb{A}\}$, and the related noncommutative Hardy space $H^r_p(\mathbb{A})$ is the completion of $H^r_\infty(\mathbb{A})$ in $L_p(\mathcal{M})$. In the present paper, we make a connection between $H^r_p(\mathbb{A})$ and certain noncommutative martingale Hardy space via showing that triangular truncation is a concrete martingale transform. We include several related results for noncommutative nest Hardy spaces in the spirit of recent development of noncommutative martingale theory.

1 Introduction

In his seminal paper [1], Arveson introduced the notion of subdiagonal algebras. Let $\mathcal{M}$ be a von Neumann algebra equipped with a normal faithful tracial state τ. Let $\mathcal{D}$ be a von Neumann subalgebra of $\mathcal{M}$, and let $\mathcal{E}:\mathcal{M}\to\mathcal{D}$ be the unique normal faithful conditional expectation such that $\tau\circ\mathcal{E}=\tau$. A weak-$*$-closed subalgebra $H_\infty(\mathcal{M})$ of $\mathcal{M}$ is called a finite subdiagonal algebra of $\mathcal{M}$ with respect to $\mathcal{E}$ (or $\mathcal{D}$) if

(i) $\{x+y^*: x,y\in H_\infty(\mathcal{M})\}$ is weak-$*$-dense in $\mathcal{M}$;
(ii) $\mathcal{E}(xy)=\mathcal{E}(x)\mathcal{E}(y)$ for all $x,y\in H_\infty(\mathcal{M})$; and
(iii) $\{x: x,x^*\in H_\infty(\mathcal{M})\}=\mathcal{D}$.

The subalgebra $\mathcal{D}$ is then called the diagonal of $H_\infty(\mathcal{M})$. According to [15], $H_\infty(\mathcal{M})$ is automatically maximal in the sense that it is contained in no proper subdiagonal algebra with respect to $\mathcal{E}$ other than itself. For $0<p<\infty$, the Hardy space $H_p(\mathcal{M})$ is defined as the closure of $H_\infty(\mathcal{M})$ in $L_p(\mathcal{M})$. If $\mathcal{M}=L_\infty(\mathbb{D})$ where $\mathbb{D}$ is the open unit disc in the complex plane, then $H_\infty(\mathcal{M})=H_\infty(\mathbb{D})$ is the algebra of analytic and bounded functions on $\mathbb{D}$ and the diagonal subalgebra $\mathcal{D}$ is given by $\mathbb{C}$; moreover, $H_p(\mathcal{M})=H_p(\mathbb{D})$ is the classical analytic Hardy space. Hence, $H_p(\mathcal{M})$'s are noncommutative extensions of classical Hardy spaces on the unit disc $\mathbb{D}$. On the other hand, the triangular algebra studied in Kadison and Singer's paper [30] (see also [37]) provides an important noncommutative example for $H_\infty(\mathcal{M})$. Many results on the classical Hardy spaces in the unit disc have been extended to the noncommutative setting. For instance, we refer the reader to [3, 8, 19, 32–34, 42, 47, 48, 53], and the survey papers [7, 45] for more information and references.

Parallel to the development of the theory of noncommutative Hardy spaces, the noncommutative martingale Hardy space theory also got rapid development recently. Indeed, in 1997, Pisier and Xu [43] introduced suitable definition of noncommmutative martingale Hardy spaces and established

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remarkable noncommutative Burkholder–Gundy inequality. Since then many classical martingale results have been successfully transferred to the noncommutative setting. For example, the duality, interpolation, atomic decomposition of noncommutative martingale Hardy spaces are investigated by many researchers; see, for instance, [4, 9, 16, 25, 26, 40, 41, 51, 52]. We also refer the reader to [22], [21], and [24] for recent advance about noncommutative martingales.

In the commutative setting, the Hardy space $H_p(\mathbb{D})$ in harmonic analysis has deep connection with martingales. For example, Maurey [35] proved the famous result that $H_1(\mathbb{D})$ is isomorphic to a dyadic martingale Hardy space. A natural question is *whether there exists some connection between the above noncommutative Hardy spaces and noncommutative martingales*. We answer this question positively in the sense of noncommutative Hardy spaces associated with nest algebras. Let $\mathbb{A}$ be a nest of order type $\mathbb{N}$, that is (see, e.g., [2, Example 2.2]),

$$\mathbb{A} = \{e_k, 1\}_{k\geq 0},$$

where 1 denotes the identity of $\mathcal{M}$, and $\{e_k\}_{k\geq 0}$ is a sequence of increasing projections in $\mathcal{M}$ such that $e_0 = 0$ and $\vee_k e_k = 1$. The nest algebra $H^r_\infty(\mathcal{M}, \mathbb{A})$ (shortly $H^r_\infty(\mathbb{A})$) with respect to $\mathbb{A}$ is defined by (see, e.g., [2,p. 11] or [12,p. 22])

$$H^r_\infty(\mathbb{A}) = \{x \in \mathcal{M} : xe = exe, \quad \forall e \in \mathbb{A}\}.$$

If $\mathcal{M} \subset B(H)$ for some Hilbert space H and each $e_k(H)$ has dimension k, then the nest algebra $H^r_\infty(\mathbb{A})$ is a triangular algebra in the sense of [30]. Moreover, the nest algebra $H^r_\infty(\mathbb{A})$ is a finite subdiagonal algebra with the diagonal algebra $\mathcal{D}$ given by $H^r_\infty(\mathbb{A}) \cap H^c_\infty(\mathbb{A})$, where $H^c_\infty(\mathbb{A}) = \{x^* : x \in H^r_\infty(\mathbb{A})\}$. For $0 < p < \infty$, the noncommutative Hardy space $H^r_p(\mathbb{A})$ is the completion of $H^r_\infty(\mathbb{A})$ with respect to the $L_p(\mathcal{M})$ norm. The symbol "r" appears in $H^r_p(\mathbb{A})$ because later we would like to compare it with row martingale Hardy spaces (see, e.g., Proposition 3.7). Note that the Hardy space $H^r_p(\mathbb{A})$ can be also defined by the triangular truncation given as follows:

$$T(x) = \sum_{k\geq 1}\sum_{j=1}^{k} \widetilde{e}_j x \widetilde{e}_k, \quad x \in H^r_\infty(\mathbb{A}),$$

where $\widetilde{e}_1 = e_1$ and $\widetilde{e}_k = e_k - e_{k-1}$, $k \geq 2$. Indeed, $x \in H^r_p(\mathbb{A})$ if and only if $x = T(x)$ and $x \in L_p(\mathcal{M})$. One of our key points for the approach is that we write the above triangular truncation T as a noncommutative martingale transform; see Proposition 3.4. This kind of phenomenon is not new. Indeed, very recently the authors of [18] investigated the relationship between main triangular truncation introduced by [31] and certain noncommutative dyadic martingale transforms.

After making the connection between noncommutative nest Hardy spaces and martingales, we study related results in the setting noncommutative nest Hardy spaces by the techniques imported from the (noncommutative) martingale theory. The main results are listed below.

In Section 3, we show that the triangular truncation T defined above is actually a concrete martingale transform (Proposition 3.4). A few results follow. For example, we show that the triangular truncation T is bounded on martingale BMO space, see Corollary 3.6. It is stronger than [46, Theorem 1]. For $0 < p \leq 1$, the Hardy space $H^T_p(\mathcal{M})$ is the sum of column and row martingale Hardy spaces $\mathcal{H}^c_p(\mathcal{M})$, $\mathcal{H}^r_p(\mathcal{M})$; see Theorem 3.11. We also prove an equivalent characterization of BMO space, see Theorem 3.16. This result is stronger than [47, Theorem 2.3] and [39, Theorem 2].

Section 4 considers noncommutative maximal inequalities with respect to the family projections $P = (P_k)_{k\geq 1}$ given in (4.1). The results in this section will be used in Section 5. Observe that Theorem 4.2 follows from the noncommutative Doob inequalities established in [11, 25]. However, we still provide another method which is based on triangular truncation. As a byproduct, we also obtain new asymmetric maximal Doob inequalities, Proposition 4.6. It has independent interest and can be compared with [17, Theorems A and B].

In Section 5, we consider atomic decomposition, duality and interpolation of the nest Hardy space $H^r_p(\mathbb{A})$. The atomic decomposition, see Theorem 5.1, is proved with the help of factorization. This result is comparable with the algebraic atomic decomposition of noncommutative martingale Hardy spaces studied in [9, 17, 50]. The duality result Theorem 5.3 is motivated by [26, Theorem 4.1]. In [26], it was

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proved that the dual space of noncommutative martingale Hardy space is some BMO type space, that is, $[\mathcal{H}_p^r(\mathcal{M})]^* = \mathcal{L}_{p'}^r\mathcal{MO}(\mathcal{M})$, $1 \le p < 2$. As a consequence of our duality result, we obtain an equivalent characterization of $H_q^r(\mathbb{A})$ with $2 < q < \infty$; see Corollary 5.6. By the duality and our characterization of BMO spaces, see Theorem 3.18, we show every upper triangular matrix in BMO space has a bounded extension, see Corollary 5.7. This is an alternative proof for [39, Theorem 1]. Finally, in spirit of Musat's [38] interpolation result for noncommutative martingales, we give a proof for the complex interpolation between $H_p^r(\mathbb{A})$ and $BMO^r(\mathbb{A})$ (see the definition after (2.6)).

2 Preliminary

Throughout this paper, denote by c a universal positive constant, which may vary from line to line, and denote by c_p a positive constant depends only on p.

2.1 Noncommutative Lebesgue and vector-valued spaces

Throughout, let $\mathcal{M} \subset B(H)$ be a finite von Neumann algebra on some Hilbert space equipped with a normalized faithful trace τ. Let $L_0(\mathcal{M})$ denote the topological $*$-algebra of all τ-measurable operators. For $0 < p < \infty$, the noncommutative Lebesgue space $L_p(\mathcal{M})$ is defined by

$$L_p(\mathcal{M}) = \{x \in L_0(\mathcal{M}) : \tau(|x|^p) < \infty\}$$

equipped with the (quasi-) norm

$$\|x\|_p = [\tau(|x|^p)]^{1/p}.$$

$L_\infty(\mathcal{M})$ is $\mathcal{M}$ equipped with the operator norm. We refer the reader to [14] for more information about the noncommutative Lebesgue spaces.

Following [25, 44], for $1 \le p \le \infty$, define $L_p(\mathcal{M}, \ell_\infty)$ to be the space of all sequences $x = (x_k)_{k\ge1}$ in $L_p(\mathcal{M})$ for which there exist $a, b \in L_{2p}(\mathcal{M})$ and a bounded sequence $y = (y_k)_{k\ge1}$ such that

$$x_k = ay_kb, \qquad k \ge 1.$$

For $x = (x_k)_{k\ge1} \in L_p(\mathcal{M}, \ell_\infty)$, define

$$\|x\|_{L_p(\mathcal{M},\ell_\infty)} = \inf\{\|a\|_{L_{2p}(\mathcal{M})} \sup_{k\ge1} \|y_k\|_\infty \|b\|_{L_{2p}(\mathcal{M})}\}, \tag{2.1}$$

where the infimum is taken over all possible factorizations of x as above. In addition, if $x = (x_k)_{k\ge1} \in E(\mathcal{M}, \ell_\infty)$ and $x_k = x_k^*$ for each $k \ge 1$, then (see [29, p. 392])

$$\|x\|_{L_p(\mathcal{M},\ell_\infty)} \le \inf\{\|a\|_{L_p(\mathcal{M})} : a \in L_p(\mathcal{M}), -a \le x_k \le a, \ \forall k \ge 1\} \le 2\|x\|_{L_p(\mathcal{M},\ell_\infty)}.$$

Following [25, 38], for $2 \le p \le \infty$, $L_p(\mathcal{M}, \ell_\infty^c)$ is defined to be the space of all sequences $x = (x_k)_{k\ge1}$ in $L_p(\mathcal{M})$ for which

$$\|(x_k)_{k\ge1}\|_{L_p(\mathcal{M},\ell_\infty^c)} = \|(|x_k|^2)_{k\ge1}\|^{1/2}_{L_{\frac{p}{2}}(\mathcal{M},\ell_\infty)} \tag{2.2}$$

is finite. For $x = (x_k)_{k\ge1}$ in $L_p(\mathcal{M})$, define

$$\|(x_k)_{k\ge1}\|_{L_p(\mathcal{M},\ell_\infty^r)} = \|(x_k^*)_{k\ge1}\|_{L_p(\mathcal{M},\ell_\infty^c)}.$$

Note that for each $1 \le p \le \infty$, $L_p(\mathcal{M}, \ell_\infty)$ is a Banach space (see [29]); and for each $2 \le p \le \infty$, $L_p(\mathcal{M}, \ell_\infty^c)$ is also a Banach space (see [38, Lemma 3.5]). In what follows, we will use the following useful lemmas.

Lemma 2.1 ([38, Lemma 3.5]). Let $2 \le p \le \infty$ and $1/p + 1/q = 1/2$. For $x = (x_k)_{k\ge1} \in L_p(\mathcal{M}, \ell_\infty^r)$, we have

$$\|x\|_{L_p(\mathcal{M},\ell_\infty^r)} = \sup\left\{\Big(\sum_{k\ge1} \|x_k^* v_k\|_2^2\Big)^{1/2} : \Big\|\Big(\sum_{k\ge1} v_k v_k^*\Big)^{1/2}\Big\|_q \le 1\right\}.$$

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Lemma 2.2 ([25, Remark 3.7]). Let $1 < p \le \infty$, and let $1/p + 1/p' = 1$. Assume that $x = (x_k)_{k\ge1}$ is sequence of positive operators in $L_p(\mathcal{M})$. We have

$$\|x\|_{L_p(\mathcal{M},\ell_\infty)} = \sup\Big\{ \sum_k \tau(x_k y_k) : y_k \ge 0, \Big\| \sum_k y_k \Big\|_{L_{p'}(\mathcal{M})} \le 1 \Big\}.$$

2.2 Nest algebra and related noncommutative Hardy and BMO spaces

Let $\mathbb{A}$ be a totally ordered family of projections in $\mathcal{M}$ containing 0 and 1. The nest algebra $H^r_\infty(\mathcal{M},\mathbb{A})$ (shortly $H^r_\infty(\mathbb{A})$) with respect to $\mathbb{A}$ is defined by

$$H^r_\infty(\mathbb{A}) = \{x \in \mathcal{M} : xe = exe, \quad \forall e \in \mathbb{A}\}.$$

Similarly, define

$$H^c_\infty(\mathbb{A}) = \{x \in \mathcal{M} : ex = exe, \quad \forall e \in \mathbb{A}\}.$$

Note that $x \in H^r_\infty(\mathbb{A})$ if and only $x^* \in H^c_\infty(\mathbb{A})$. For $0 < p < \infty$, the noncommutative Hardy space $H^r_p(\mathbb{A})$ (resp. $H^c_p(\mathbb{A})$) is the completion of $H^r_\infty(\mathbb{A})$ (resp. $H^c_\infty(\mathbb{A})$) with respect to the $L_p(\mathcal{M})$ norm.

Remark 2.3. In this paper, we only consider the nest of order type $\mathbb{N}$, that is,

$$\mathbb{A} = \{e_k, 1\}_{k\ge0},$$

where $\{e_k\}_{k\ge0}$ is a sequence of increasing projections in $\mathcal{M}$ such that $e_0 = 0$ and $\vee_k e_k = 1$. We refer the reader to [2, Example 2.2] for more details.

The following result, taken from [3, Theorem 3.4], is the well-known Riesz factorization theorem of Hardy spaces.

Theorem 2.4. Let $0 < p, q, s \le \infty$ with $1/s = 1/p + 1/q$. Then, each $x \in H^r_s(\mathbb{A})$ can be factored as $x = yz$ with $y = H^r_p(\mathbb{A})$ and $z \in H^r_q(\mathbb{A})$; moreover,

$$\|x\|_s = \inf\{\|y\|_p \|z\|_q : x = yz, y \in H^r_p(\mathbb{A}), z \in H^r_q(\mathbb{A})\}.$$

The same result holds for the $H^c_s(\mathbb{A})$ spaces.

We also introduce BMO type spaces as follows. For each $k \ge 1$, let

$$\mathcal{Z}_{r,k}(x) = e_k x(1 - e_{k-1}), \quad \forall x \in \mathcal{M}. \tag{2.3}$$

and

$$\mathcal{Z}_{c,k}(x) = (1 - e_{k-1})xe_k, \quad \forall x \in \mathcal{M}. \tag{2.4}$$

For $2 \le q \le \infty$, define

$$L^r_q MO(\mathbb{A}) = \{x \in H^r_\infty(\mathbb{A}) : \|x\|_{L^r_q MO} := \|(\mathcal{Z}_{r,k}(x))_{k\ge1}\|_{L_q(\mathcal{M},\ell^r_\infty)} < \infty\} \tag{2.5}$$

and

$$L^c_q MO(\mathbb{A}) = \{x \in H^c_\infty(\mathbb{A}) : \|x\|_{L^c_q MO} := \|x^*\|_{L^r_q MO} < \infty\}. \tag{2.6}$$

In the case when $q = \infty$, we write $BMO^r(\mathbb{A})$ (resp. $BMO^c(\mathbb{A})$) instead of $L^r_\infty MO(\mathbb{A})$ (resp. $L^c_\infty MO(\mathbb{A})$).

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2.3 Noncommutative martingales and related Hardy and BMO spaces

Let $(\mathcal{M}_n)_{n\geq 1}$ be a sequence of increasing von Neumann sub-algebras of $\mathcal{M}$ satisfying $\cup_n \mathcal{M}_n$ is weak* dense in $\mathcal{M}$. Assume that for each $n \geq 1$ there exists conditional expectation $\mathcal{E}_n : \mathcal{M} \to \mathcal{M}_n$.

The conditional expectations $(\mathcal{E}_n)_{n\geq 1}$ satisfy the following:

(a) $\mathcal{E}_n(xy) = \mathcal{E}_n(x)y$, $\mathcal{E}_n(yx) = y\mathcal{E}_n(x)$, $n \geq 1$ and $y \in \mathcal{M}_n$;
(b) $\mathcal{E}_n\mathcal{E}_m = \mathcal{E}_n$ for $m \geq n$; and
(c) $\tau(\mathcal{E}_n(x)) = \tau(x)$, $n \geq 1$.

Since each $\mathcal{E}_n$ preserves the trace, it extends to a contractive projection from $L_p(\mathcal{M}, \tau)$ onto $L_p(\mathcal{M}_n, \tau)$ for all $1 \leq p \leq \infty$.

A sequence $x = (x_n)_{n\geq 1} \subset L_1(\mathcal{M})$ is called a martingale if

$$\mathcal{E}_n(x_{n+1}) = x_n, \quad n \geq 1.$$

If in addition, $(x_n)_{n\geq 1} \subset L_p(\mathcal{M})$ with $1 \leq p < \infty$, then x is called an L_p-martingale. In this case, set

$$\|x\|_p = \sup_{n\geq 1} \|x_n\|_p.$$

If $\|x\|_p < \infty$, then $x = (x_n)_{n\geq 1}$ is called a bounded L_p-martingale. A martingale is called finite if there exists some N such that $x_m = x_N$ for $m \geq N$. In what follows, the sequence of martingale differences corresponding to the martingale $x = (x_n)_{n\geq 1}$ is denoted by $(d_n x)_{n\geq 1}$ with $d_1 x = x_1$ and

$$d_n x = x_n - x_{n-1}, \quad \forall n \geq 2. \tag{2.7}$$

For an element $y \in L_1(\mathcal{M})$, note that the sequence $(\mathcal{E}_n(y))_{n\geq 1}$ is a martingale.

Remark 2.5. If $x \in L_p(\mathcal{M})$ with $1 \leq p < \infty$, then $(\mathcal{E}_n(x))_{n\geq 1}$ is a bounded L_p-martingale and $\mathcal{E}_n(x)$ converges to x in $L_p(\mathcal{M})$. Conversely, if $(x_n)_{n\geq 1}$ is a bounded L_p-martingale with $1 < p < \infty$, then there exists $x_\infty \in L_p(\mathcal{M})$ such that $x_n = \mathcal{E}_n(x_\infty)$ for each $n \geq 1$. In this paper, we often identify a martingale with its final value x_∞ whenever the latter exists.

For $0 < p < \infty$ and any finite sequence $a = (a_n)_{n\geq 1} \subset L_p(\mathcal{M})$, define

$$\|a\|_{L_p(\mathcal{M},\ell_2^c)} = \Big\| \Big(\sum_{n\geq 1} |a_n|^2 \Big)^{1/2} \Big\|_p, \quad \|a\|_{L_p(\mathcal{M},\ell_2^r)} = \Big\| \Big(\sum_{n\geq 1} |a_n^*|^2 \Big)^{1/2} \Big\|_p.$$

This gives two quasi-norms on the family of all finite sequences in $L_p(\mathcal{M})$. The corresponding completion is a Banach space, denoted by $L_p(\mathcal{M}, \ell_2^c)$ or $L_p(\mathcal{M}, \ell_2^r)$, respectively. It is easy to check that a sequence $a = (a_n)_{n\geq 1} \subset L_p(\mathcal{M})$ belong to $L_p(\mathcal{M}, \ell_2^c)$ if and only if $\sup_N \|(\sum_{n=1}^N |a_n|^2)^{1/2}\|_p < \infty$, if this is the case, then $(\sum_{n=1}^N |a_n|^2)^{1/2}$ converges to $(\sum_{n\geq 1} |a_n|^2)^{1/2}$ in $L_p(\mathcal{M})$.

Recall that for $1 \leq p < \infty$ $L_p(\mathcal{M}, \ell_2^c)$ (resp. $L_p(\mathcal{M}, \ell_2^r)$) is viewed 1-complemented subspace of $L_p(\mathcal{M}\bar{\otimes}B(\ell_2))$; see, for example, [43,p. 670]. Hence, the following duality can be deduced from the duality between $L_p(\mathcal{M}\bar{\otimes}B(\ell_2))$ and $L_q(\mathcal{M}\bar{\otimes}B(\ell_2))$ with $1 \leq p < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$:

$$(L_p(\mathcal{M}, \ell_2^c))^* = L_q(\mathcal{M}, \ell_2^c), \quad (L_p(\mathcal{M}, \ell_2^r))^* = L_q(\mathcal{M}, \ell_2^r). \tag{2.8}$$

For $0 < p \leq 2$, define $\mathcal{H}_p^r(\mathcal{M})$ (resp. $\mathcal{H}_p^c(\mathcal{M})$) as the completion of all finite L_2-martingales under the quasi-norm

$$\|x\|_{\mathcal{H}_p^r} := \|(d_n x)_{n\geq 1}\|_{L_p(\mathcal{M},\ell_2^r)} \quad (\text{resp. } \|x\|_{\mathcal{H}_p^c} := \|(d_n x)_{n\geq 1}\|_{L_p(\mathcal{M},\ell_2^c)}).$$

If $x \in \mathcal{H}_p^c(\mathcal{M})$ or $x \in \mathcal{H}_p^r(\mathcal{M})$, then $S_c(x) \in L_p(\mathcal{M})$ or $S_r(x) \in L_p(\mathcal{M})$, where the column and row square functions $S_c(x)$ and $S_r(x)$ are defined by

$$S_c(x) = \Big(\sum_{n\geq 1} |d_n x|^2 \Big)^{1/2}, \quad S_r(x) = \Big(\sum_{n\geq 1} |d_n x^*|^2 \Big)^{1/2}. \tag{2.9}$$

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For $2 \le q \le \infty$, define (let $\mathcal{E}_0 = 0$ for convenience)

$$\mathcal{L}_q^r\mathcal{MO}(\mathcal{M}) = \{x \in L_2(\mathcal{M}) : \|(\mathcal{E}_n|x^* - \mathcal{E}_{n-1}(x^*)|^2)_{n\ge1}\|_{L_{\frac{q}{2}}(\mathcal{M},\ell_\infty)} < \infty\}$$

equipped with the norm

$$\|x\|_{\mathcal{L}_q^r\mathcal{MO}} = \|(\mathcal{E}_n|x^* - \mathcal{E}_{n-1}(x^*)|^2)_{n\ge1}\|_{L_{\frac{q}{2}}(\mathcal{M},\ell_\infty)}^{\frac{1}{2}}.$$

Similarly, define

$$\mathcal{L}_q^c\mathcal{MO}(\mathcal{M}) = \{x : x^* \in \mathcal{L}_q^r\mathcal{MO}(\mathcal{M})\}$$

equipped with the norm

$$\|x\|_{\mathcal{L}_q^c\mathcal{MO}} = \|x^*\|_{\mathcal{L}_q^r\mathcal{MO}}.$$

In the case when $q = \infty$, we write $\mathcal{BMO}^c(\mathcal{M})$ (resp. $\mathcal{BMO}^r(\mathcal{M})$) instead of $\mathcal{L}_\infty^c\mathcal{MO}(\mathcal{M})$ (resp. $\mathcal{L}_\infty^c\mathcal{MO}(\mathcal{M})$). These are noncommutative martingale BMO spaces (see, e.g., [43]).

The following filtration is associated with the growing family of projections $\mathbb{A}$ and is called the shell filtration. A number of the results that we present below are proved for the shell filtration. We clearly state that each time. We do not know if those results work for other filtrations.

Example 2.6. Let $\mathbb{A}$ be as in Remark 2.3. Set

$$\widetilde{e}_1 = e_1, \quad \widetilde{e}_k = e_k - e_{k-1}, \quad k \ge 2. \tag{2.10}$$

Then $(\widetilde{e}_k)_{k\ge1}$ are mutually orthogonal. Let $k \ge 1$, and let

$$M_k = \{a : a \in e_k\mathcal{M}e_k \quad \text{or} \quad a \in \bigcup_{n>k} \widetilde{e}_n\mathcal{M}\widetilde{e}_n\}, \quad \mathcal{M}_k = \mathrm{VNA}(M_k), \tag{2.11}$$

where $\mathrm{VNA}(M_k)$ denotes the von Neumann algebra generated by M_k. Then $(\mathcal{M}_k)_{k\ge1}$ is a sequence of increasing von Neumann subalgebras of $\mathcal{M}$. Moreover, the map $\mathcal{E}_k : \mathcal{M} \to \mathcal{M}_k$ defined by

$$\mathcal{E}_k(x) = e_kxe_k + \sum_{n>k} \widetilde{e}_nx\widetilde{e}_n \tag{2.12}$$

is the corresponding conditional expectation. The sum makes sense for the strong operator topology; see for instance [12, Proposition 4.4].

2.4 Useful lemmas

We collect several useful lemmas in this subsection.

The following lemma is well known, and we refer the reader to [10, Section 3] and [6] for more information.

Lemma 2.7. Assume that $(q_k)_{k\ge1} \subset \mathcal{M}$ are mutually orthogonal projections. For each $x \in L_p(\mathcal{M})$, we have

$$\|B(x)\|_{L_p(\mathcal{M})} \le \|x\|_{L_p(\mathcal{M})}, \quad 1 \le p \le \infty,$$

where B is the block projection defined by $B(x) = \sum_{k\ge1} q_kxq_k$.

The following basic lemma is taken from [36, Proposition 1.1].

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Lemma 2.8. Let $0 < p, q, r \le \infty$ with $1/p + 1/q = 1/r$. Assume that finite sequences $(a_k)_k \subset L_p(\mathcal{M})$ and $(b_k)_k \subset L_q(\mathcal{M})$. We have

$$\Big\| \sum_k a_k^* b_k \Big\|_r \le \Big\| \Big(\sum_k |a_k|^2 \Big)^{1/2} \Big\|_p \Big\| \Big(\sum_k |b_k|^2 \Big)^{1/2} \Big\|_q .$$

3 Triangular Truncation and Martingale Transforms

In this section, first we show that each triangular truncation is seen as a martingale transform. Then, we explore the connection between the Hardy space $H_p^r(\mathbb{A})$ (resp. $BMO^r(\mathbb{A})$) and the martingale Hardy space $\mathcal{H}_p^c(\mathcal{M})$ (resp. martingale BMO space $\mathcal{BMO}^r(\mathcal{M})$) with respect to the shell filtration $(\mathcal{M}_k)_{k\ge1}$ given in (2.11). We always assume that each martingale has the form $x = (\mathcal{E}_n(x))_{n\ge1}$, $x \in L_1(\mathcal{M})$ (see Remark 2.5).

3.1 Triangular truncation and martingale transform

Let $\mathbf{e} = (\widetilde{e}_k)_k$ be as in (2.10). The triangular truncation operators T and $T^\perp$ with respect to $\mathbf{e}$ are defined by

$$T(x) = \sum_{k\ge1} \sum_{j=1}^{k} \widetilde{e}_j x \widetilde{e}_k, \quad T^\perp(x) = x - T(x), \quad x \in L_2(\mathcal{M}). \tag{3.1}$$

Observe that $T(x)$ is well defined. Indeed, writing $T_N(x) = \sum_{k=1}^N \sum_{j=1}^k \widetilde{e}_j x \widetilde{e}_k$, $T(x) = \lim_{N\to\infty} T_N(x)$ in $L_2(\mathcal{M})$.

Recall that triangular truncation operators are of strong type (p,p) with $1 < p < \infty$, and of weak type $(1,1)$; see, for instance, [13, Theorem 1.4].

Lemma 3.1. We have

$$\|T(x)\|_{L_p(\mathcal{M})}, \quad \|T^\perp(x)\|_{L_p(\mathcal{M})} \le c_p \|x\|_{L_p(\mathcal{M})}, \quad 1 < p < \infty,$$

and

$$\|T(x)\|_{L_{1,\infty}(\mathcal{M})}, \quad \|T^\perp(x)\|_{L_{1,\infty}(\mathcal{M})} \le c \|x\|_{L_1(\mathcal{M})}.$$

Here $c_p = O(1/(p-1))$ as $p \to 1$.

Next we study the relationship between triangular truncation operators and martingale transforms. To begin with, we show that the shell filtration $(\mathcal{M}_k)_{k\ge1}$, defined in Example 2.6, is regular. Recall that we say a filtration is regular if for any positive $x \in \mathcal{M}$, there exists a positive constant c such that

$$\mathcal{E}_k(x) \le c\mathcal{E}_{k-1}(x).$$

Lemma 3.2. The shell filtration $(\mathcal{M}_k)_{k\ge1}$ given in (2.11) is regular with constant 2.

Proof. Take positive $x \in \mathcal{M}$. We shall show

$$\mathcal{E}_k(x) \le 2\mathcal{E}_{k-1}(x), \quad k \ge 2.$$

We start with

$$(e_{k-1} - \widetilde{e}_k) x (e_{k-1} - \widetilde{e}_k) \ge 0.$$

Opening the brackets, we arrive at

$$\widetilde{e}_k x e_{k-1} + e_{k-1} x \widetilde{e}_k \le e_{k-1} x e_{k-1} + \widetilde{e}_k x \widetilde{e}_k.$$

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Hence, for each k,

$$\begin{aligned} e_k x e_k &= (\widetilde{e}_k + e_{k-1})x(\widetilde{e}_k + e_{k-1}) \\ &= \widetilde{e}_k x e_{k-1} + e_{k-1} x \widetilde{e}_k + e_{k-1} x e_{k-1} + \widetilde{e}_k x \widetilde{e}_k \\ &\le 2\Big(e_{k-1} x e_{k-1} + \widetilde{e}_k x \widetilde{e}_k\Big). \end{aligned} \tag{3.2}$$

Furthermore, according to (2.12),

$$\begin{aligned} \mathcal{E}_k(x) &= e_k x e_k + \sum_{m \ge k+1} \widetilde{e}_m x \widetilde{e}_m \\ &\le 2\Big(e_{k-1} x e_{k-1} + \widetilde{e}_k x \widetilde{e}_k + \sum_{m \ge k+1} \widetilde{e}_m x \widetilde{e}_m\Big) \\ &= 2\mathcal{E}_{k-1}(x). \end{aligned}$$

The proof is complete. ■

Recall that the martingale difference operator is denoted by $d_j = \mathcal{E}_j - \mathcal{E}_{j-1}$ for each $j \ge 1$ (let $\mathcal{E}_0 = 0$ for convenience).

Lemma 3.3. For each $k \ge 2$ and $x \in \mathcal{M}$, we have

$$e_{k-1} x \widetilde{e}_k = e_{k-1} d_k(x).$$

Similarly,

$$\widetilde{e}_k x e_{k-1} = d_k(x) e_{k-1}.$$

Proof. By (2.12), for each $j \ge 2$, we have

$$d_j(x) = e_j x e_j - e_{j-1} x e_{j-1} - \widetilde{e}_j x \widetilde{e}_j$$

We write

$$e_j x e_j = (\widetilde{e}_j + e_{j-1})x(\widetilde{e}_j + e_{j-1}).$$

Then, combining the last two formulas, we have

$$d_j(x) = \widetilde{e}_j x e_{j-1} + e_{j-1} x \widetilde{e}_j, \quad j \ge 2. \tag{3.3}$$

Thus, for each $k \ge 2$, we have

$$e_{k-1} d_k(x) = e_{k-1}(\widetilde{e}_k x e_{k-1} + e_{k-1} x \widetilde{e}_k) = e_{k-1} x \widetilde{e}_k.$$

The second equality can be similarly proved. ■

Now we are ready to show that triangular truncation is a martingale transform (see Remark 3.5 below for detail explanation).

Proposition 3.4. Let T and $T^{\perp}$ be the triangular truncation defined in (3.1). For each $x \in L_2(\mathcal{M})$, we have $\mathcal{E}_1(T(x)) = \mathcal{E}_1(x)$,

$$d_n(T(x)) = e_{n-1} d_n(x), \quad n \ge 2, \tag{3.4}$$

and consequently,

$$T(x) = \mathcal{E}_1(x) + \sum_{k \ge 2} e_{k-1} d_k(x).$$

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Similarly, we have

$$T^{\perp}(x) = \sum_{k\geq 2} d_k(x)e_{k-1}, \quad x \in L_2(\mathcal{M}).$$

Proof. Let $x \in L_2(\mathcal{M})$. For each $N \geq 2$, we have

$$T_N(x) = \sum_{k=1}^{N}\sum_{j=1}^{k} \widetilde{e}_j x \widetilde{e}_k.$$

Note that $T_N(x)$ converges to $T(x)$ in $L_2(\mathcal{M})$ as $N \to \infty$. Hence, for each $n \geq 1$,

$$\lim_{N\to\infty} \|\mathcal{E}_k(T_N(x) - T(x))\|_{L_2(\mathcal{M})} \leq \lim_{N\to\infty} \|T_N(x) - T(x)\|_{L_2(\mathcal{M})} = 0.$$

Consequently,

$$\mathcal{E}_1(T(x)) = \lim_{N\to\infty} \mathcal{E}_1(T_N(x)) \quad \text{in } L_2(\mathcal{M}) \tag{3.5}$$

and for $n \geq 2$

$$d_n(T(x)) = \lim_{N\to\infty} d_n(T_N(x)) \quad \text{in } L_2(\mathcal{M}). \tag{3.6}$$

For each N, it follows from (2.12) that $\mathcal{E}_1(T_N(x)) = \sum_{k=1}^{N} \widetilde{e}_k x \widetilde{e}_k$. By Lemma 2.7, we have

$$\sum_{k=1}^{N} \widetilde{e}_k x \widetilde{e}_k \to \sum_{k\geq 1} \widetilde{e}_k x \widetilde{e}_k = \mathcal{E}_1(x), \quad N \to \infty$$

in $L_2(\mathcal{M})$. Thus, it follows from (3.5) that $\mathcal{E}_1(T(x)) = \mathcal{E}_1(x)$. On the other hand, for $N \geq n$, by (2.12), we have

$$\mathcal{E}_n(T_N(x)) = \sum_{k=1}^{N} \widetilde{e}_k x \widetilde{e}_k + \sum_{k=2}^{n} e_{k-1} x \widetilde{e}_k.$$

Therefore, for $N \geq n$,

$$d_n(T_N(x)) = \mathcal{E}_n(T_N(x)) - \mathcal{E}_{n-1}(T_N(x)) = e_{n-1} x \widetilde{e}_n = e_{n-1} d_n(x),$$

where the last equality is due to Lemma 3.3. From (3.6), we deduce that $d_n(T(x)) = e_{n-1}d_n(x)$. Note that

$$0 = \lim_{N\to\infty} \|T(x) - \mathcal{E}_N(T(x))\|_{L_2(\mathcal{M})} = \lim_{N\to\infty} \|T(x) - [\mathcal{E}_1(T(x)) + \sum_{k=1}^{N} d_k(T(x))]\|_{L_2(\mathcal{M})}.$$

Thus, we conclude from (3.4) that

$$T(x) = \mathcal{E}_1(x) + \sum_{k\geq 2} e_{k-1} d_k(x).$$

The formula for $T^{\perp}$ can be similarly proved. ∎

Remark 3.5. For a given bounded L_1-martingale $x = (x_n)_{n\geq 1}$, define a martingale y by setting

$$d_n y = \xi_{n-1} d_k(x), \quad \xi_{n-1} \in \mathcal{M}_{n-1}, \quad n \geq 1.$$

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We call y is a martingale transform of x (see, for instance, [49], [40]). Note that $e_{k-1} \in \mathcal{M}_{k-1}$ for each k. Hence, the above result actually states that $T(x)$ is a martingale transform of $x = (\mathcal{E}_n(x))_{n\geq 1}$. Motivated by this, in the forthcoming paper, we will provide a new martingale proof for the weak $(1,1)$ boundedness of the triangular truncation T.

It was proved in [46, Theorem 1] that for each $x \in \mathcal{M}$, there exists a positive constant c such that

$$\max\{\|T(x)\|_{\mathcal{BMO}^c(\mathcal{M})}, \|T(x)\|_{\mathcal{BMO}^r(\mathcal{M})}\} \leq c\|x\|_{\mathcal{M}}.$$

We improve this result with the help of Proposition 3.4. To see the improvement, one needs to recall that $\max\{\|x\|_{\mathcal{BMO}^c(\mathcal{M})}, \|x\|_{\mathcal{BMO}^r(\mathcal{M})}\} \leq 2\|x\|_{\mathcal{M}}$ for each $x \in \mathcal{M}$ (see, e.g., [43, p. 693]).

Corollary 3.6. For each $x \in \mathcal{BMO}^c(\mathcal{M})$, we have

$$\|T(x)\|_{\mathcal{BMO}^c(\mathcal{M})} \leq \|x\|_{\mathcal{BMO}^c(\mathcal{M})}$$

and

$$\|T(x)\|_{\mathcal{BMO}^r(\mathcal{M})} \leq 2\|x\|_{\mathcal{BMO}^r(\mathcal{M})}.$$

Proof. It is known that (see, e.g., [43, p. 693])

$$\mathcal{E}_n(|y-\mathcal{E}_{n-1}(y)|^2) = \mathcal{E}_n\Big(\sum_{k\geq n}|d_k y|^2\Big), \quad y \in L_2(\mathcal{M}).$$

Hence, from this and the definition of $\mathcal{BMO}^c(\mathcal{M})$, we have

$$\begin{aligned}
\|T(x)\|_{\mathcal{BMO}^c(\mathcal{M})} &= \sup_n \Big\|\mathcal{E}_n\Big(\sum_{k\geq n}|d_k(T(x))|^2\Big)\Big\|_\infty^{1/2} \\
&= \sup_n \Big\|\mathcal{E}_n\Big(\sum_{k\geq n}|e_{k-1}d_k(x)|^2\Big)\Big\|_\infty^{1/2} \\
&\leq \sup_n \Big\|\mathcal{E}_n\Big(\sum_{k\geq n}|d_k(x)|^2\Big)\Big\|_\infty^{1/2} \\
&= \|x\|_{\mathcal{BMO}^c(\mathcal{M})},
\end{aligned}$$

where we used Proposition 3.4 in the second equality and the "$\leq$" is due to the fact $e_{k-1} \leq 1$. This shows the first inequality. With the same argument, applying 3.4 for $T^\perp$, we obtain

$$\|T^\perp(x)\|_{\mathcal{BMO}^r(\mathcal{M})} \leq \|x\|_{\mathcal{BMO}^r(\mathcal{M})}.$$

From this, we have

$$\|T(x)\|_{\mathcal{BMO}^r(\mathcal{M})} = \|x - T^\perp(x)\|_{\mathcal{BMO}^r(\mathcal{M})} \leq \|x\|_{\mathcal{BMO}^r(\mathcal{M})} + \|T^\perp(x)\|_{\mathcal{BMO}^r(\mathcal{M})} \leq 2\|x\|_{\mathcal{BMO}^r(\mathcal{M})}.$$

This is the second desired inequality. ∎

3.2 Comparison: Hardy spaces and martingale Hardy spaces

Let us now compare the Hardy space $H_p^r(\mathbb{A})$ defined with the help of sequence of projections and the martingale Hardy space $\mathcal{H}_p^c(\mathcal{M})$ with respect to the shell filtration $(\mathcal{M}_k)_{k\geq 1}$ given in (2.11). The first result is as follows.

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Proposition 3.7. For $0 < p \le \infty$, we have

$$H_p^r(\mathbb{A}) \subset \mathcal{H}_p^r(\mathcal{M}).$$

Similarly, we have $H_p^c(\mathbb{A}) \subset \mathcal{H}_p^c(\mathcal{M})$.

We need the following nontrivial result from [3, Theorem 2.1].

Lemma 3.8. If $x \in H_\infty^r(\mathbb{A})$, then

$$\|\mathcal{E}_1(x)\|_p \le \|x\|_p, \quad 0 < p < 1.$$

Now we prove Proposition 3.7.

Proof of Proposition 3.7. It suffices to prove that for each $x \in H_\infty^r(\mathbb{A})$,

$$\|x\|_{\mathcal{H}_p^r} \le c_p \|x\|_p, \quad 0 < p \le \infty.$$

Note that in this case $xe_k = e_k x e_k$ for all $k \ge 1$. Hence, by (3.3), we have

$$d_j(x) = \widetilde{e}_j x e_{j-1} + e_{j-1} x \widetilde{e}_j = \widetilde{e}_j e_{j-1} x e_{j-1} + e_{j-1} x \widetilde{e}_j = e_{j-1} x \widetilde{e}_j, \quad j \ge 2,$$

and consequently,

$$x = \mathcal{E}_1(x) + \sum_{j\ge 2} d_j(x) = \mathcal{E}_1(x) + \sum_{j\ge 2} e_{j-1} x \widetilde{e}_j.$$

Writing $y = x - \mathcal{E}_1(x)$, we have $d_j(y) = d_j(x) = e_{j-1} x \widetilde{e}_j$ and

$$S_r(y) = \Big(\sum_{j\ge 2} d_j(y) d_j(y)^*\Big)^{1/2} = (yy^*)^{1/2}. \tag{3.7}$$

By Lemma 2.7 and Lemma 3.8,

$$\|\mathcal{E}_1(x)\|_p = \Big\|\sum_{k\ge 1} \widetilde{e}_k x \widetilde{e}_k\Big\|_p \le \|x\|_p, \quad 0 < p \le \infty. \tag{3.8}$$

Using the fact that $\|\cdot\|_{\mathcal{H}_p^r}$ is a quasi-norm, we have

$$\begin{aligned}
\|x\|_{\mathcal{H}_p^r} &\le c_p \|y\|_{\mathcal{H}_p^r} + c_p \|\mathcal{E}_1(x)\|_{\mathcal{H}_p^r} \\
&= c_p \Big\|\big(\sum_{j\ge 2} d_j(y) d_j(y)^*\big)^{1/2}\Big\|_p + c_p \|\mathcal{E}_1(x)\|_p \\
&\overset{(3.7)}{=} c_p \|(yy^*)^{1/2}\|_p + c_p \|\mathcal{E}_1(x)\|_p \\
&= c_p \|y\|_p + c_p \|\mathcal{E}_1(x)\|_p \\
&\le c_p^2 \|x\|_p + c_p^2 \|\mathcal{E}_1(x)\|_p + c_p \|\mathcal{E}_1(x)\|_p \overset{(3.8)}{\le} c_p \|x\|_p.
\end{aligned}$$

The proof is complete. ∎

Before going further, let us state the following result which will be used to prove Theorem 3.11. Actually, the following martingale inequalities hold for general regular noncommutative martingales.

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Lemma 3.9. For $0 < p \le 2$ and $x \in L_2(\mathcal{M})$, we have

$$\|x\|_p \le c_p \|x\|_{\mathcal{H}_p^c}, \quad \|x\|_p \le c_p \|x\|_{\mathcal{H}_p^r}.$$

Proof. The case $1 < p < 2$ follows from the noncommutative Burkholder–Gundy inequality [43, Theorem 2.1], and the case $p = 1$ is given in [43,p. 696].

Now we consider the case $0 < p < 1$. According to [20, Theorem 4.11], we have

$$\sup_n \|\mathcal{E}_n(x)\|_p \le c_p \|x\|_{\mathrm{h}_p^c}.$$

Here h_p^c denotes the conditioned martingale Hardy space (see [9] or [20] for more information). Since the shell filtration $(\mathcal{M}_k)_{k\ge1}$ is regular (see Lemma 3.2 above), it follows from [9, Theorem 5.9] that $\mathrm{h}_p^c = \mathcal{H}_p^c$ with equivalent quasi-norms. Thus,

$$\|x\|_p \le \sup_n \|\mathcal{E}_n(x)\|_p \le c_p \|x\|_{\mathcal{H}_p^c},$$

where the first inequality is due to the fact $\|x\|_p \le \|\mathcal{E}_n(x)\|_p$, $n \ge 1$, provided $0 < p < 1$ (see, e.g., [26, Theorem 7.1]). The second inequality of the lemma can be proved similarly. ■

Proposition 3.10. For each $x \in L_2(\mathcal{M})$, there exists a positive constant c_p such that

$$\|T(x)\|_p \le c_p \|x\|_{\mathcal{H}_p^c}, \quad 0 < p \le 2 \tag{3.9}$$

and

$$\|T(x)\|_p \le c_p \|x\|_{\mathcal{H}_p^r}, \quad 0 < p \le 2. \tag{3.10}$$

Similarly, we have

$$\|T^{\perp}(x)\|_p \le c_p \min\{\|x\|_{\mathcal{H}_p^r}, \|x\|_{\mathcal{H}_p^c}\}, \quad 0 < p \le 2.$$

Proof. We first show (3.9). The case when $p = 2$ is trivial. Hence, we only need to consider $0 < p < 2$. By Lemma 3.9 and Proposition 3.4, we have

$$\begin{aligned}
\|T(x)\|_p &\le c_p \|T(x)\|_{\mathcal{H}_p^c} \\
&= \left\| \left(|\mathcal{E}_1(x)|^2 + \sum_{k\ge2} |e_{k-1} d_k x|^2 \right)^{1/2} \right\|_p \\
&\le \left\| \left(|\mathcal{E}_1(x)|^2 + \sum_{k\ge2} |d_k x|^2 \right)^{1/2} \right\|_p = \|x\|_{\mathcal{H}_p^c}.
\end{aligned}$$

By the same argument, we show that

$$\|T^{\perp}(x)\|_p \le c_p \|x\|_{\mathcal{H}_p^r}. \tag{3.11}$$

Now we turn to deal with (3.10). Combining (3.11) and Lemma 3.9, we write

$$\begin{aligned}
\|T(x)\|_p &\le c_p \|T^{\perp}(x)\|_p + c_p \|x\|_p \\
&\le c_p \|x\|_{\mathcal{H}_p^r} + c_p \|x\|_{\mathcal{H}_p^r} \le c_p \|x\|_{\mathcal{H}_p^r}.
\end{aligned}$$

The proof is complete. ■

For the triangular truncation T and $0 < p \le 2$, let us define the space $H_p^T(\mathcal{M})$ as the completion of $L_2(\mathcal{M})$ with respect to the quasi-norm

$$\|x\|_{H_p^T} = \|x\|_p + \|T(x)\|_p.$$

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For each $x \in L_2(\mathcal{M})$ and $1 < p \le 2$, by Lemma 3.1, we have $\|x\|_p \le \|x\|_{H_p^T} \le (c_p+1)\|x\|_p$. Hence, for $1 < p \le 2$, $L_p(\mathcal{M}) = H_p^T(\mathcal{M})$ with equivalent norms. Also recall that (see [43, Corollary 2.2])

$$L_p(\mathcal{M}) = \mathcal{H}_p^c(\mathcal{M}) + \mathcal{H}_p^r(\mathcal{M}), \quad 1 < p \le 2,$$

with equivalent norms. Consequently,

$$H_p^T(\mathcal{M}) = \mathcal{H}_p^c(\mathcal{M}) + \mathcal{H}_p^r(\mathcal{M}), \quad 1 < p \le 2$$

with equivalent norms. We shall show below that this also holds for $0 < p \le 1$.

Theorem 3.11. For $0 < p \le 2$, we have

$$H_p^T(\mathcal{M}) = \mathcal{H}_p^c(\mathcal{M}) + \mathcal{H}_p^r(\mathcal{M})$$

with equivalent quasi-norms.

Proof. It suffices to show that for each $x \in L_2(\mathcal{M})$,

$$\|x\|_{H_p^T} \le c_p \|x\|_{\mathcal{H}_p^c + \mathcal{H}_p^r}$$

and

$$\|x\|_{\mathcal{H}_p^c + \mathcal{H}_p^r} \le c_p \|x\|_{H_p^T}. \tag{3.12}$$

The first inequality above easily follows from Lemma 3.9 and Proposition 3.10.

It remains to show (3.12). To this end, write $x = T(x) + T^{\perp}(x)$. Then

$$\begin{aligned}
\|x\|_{\mathcal{H}_p^c + \mathcal{H}_p^r} &\le \|T(x)\|_{\mathcal{H}_p^r} + \|T^{\perp}(x)\|_{\mathcal{H}_p^c} \\
&\le c_p \|T(x)\|_p + c_p \|T^{\perp}(x)\|_p \\
&\le c_p \|T(x)\|_p + c_p [c_p (\|x\|_p + \|T(x)\|_p)], \\
&= c_p (\|x\|_p + \|T(x)\|_p) = c_p \|x\|_{H_p^T},
\end{aligned}$$

where the second inequality is due to Proposition 3.7. The proof of the theorem is complete. ■

3.3 BMO type spaces

In this subsection, we aim to compare different BMO type spaces. We begin with the following basic lemma. Unless otherwise specified, the results of this subsection are stated for the shell filtration.

Lemma 3.12. For each $x \in L_2(\mathcal{M})$ and $n \ge 1$,

$$\mathcal{E}_n(|x - \mathcal{E}_{n-1}(x)|^2) = e_{n-1}x^*\widetilde{e}_n x e_{n-1} + e_n x^*(1-e_n)xe_n + \sum_{k \ge n+1} \widetilde{e}_k x^*(1-e_k)x\widetilde{e}_k + \sum_{j \ge n} \widetilde{e}_j x^* e_{j-1} x \widetilde{e}_j.$$

Moreover, if $e_k x = e_k x e_k$ for each $k \ge 1$, then for each $n \ge 1$

$$\mathcal{E}_n(|x - \mathcal{E}_{n-1}(x)|^2) = e_{n-1}x^*\widetilde{e}_n x e_{n-1} + e_n x^*(1-e_n)xe_n + \sum_{k \ge n+1} \widetilde{e}_k x^*(1-e_k)x\widetilde{e}_k. \tag{3.13}$$

Proof. For $x \in L_2(\mathcal{M})$, by the orthogonality of martingale differences, we have (see also [43,p. 693])

$$\mathcal{E}_n(|x - \mathcal{E}_{n-1}(x)|^2) = \sum_{k \ge n} \mathcal{E}_n(|d_k(x)|^2) = |d_n(x)|^2 + \sum_{j \ge n+1} \mathcal{E}_n(|d_j(x)|^2). \tag{3.14}$$

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Here the sum makes sense in $L_2(\mathcal{M})$. Note that by (3.3) we have

$$|d_j(x)|^2 = e_{j-1}x^*\widetilde{e}_j x e_{j-1} + \widetilde{e}_j x^* e_{j-1} x \widetilde{e}_j, \quad j \geq 1. \tag{3.15}$$

According to (2.12), we have, for $j \geq n+1$

$$\begin{aligned}\mathcal{E}_n(e_{j-1}x^*\widetilde{e}_j x e_{j-1}) &= e_n e_{j-1}x^*\widetilde{e}_j x e_{j-1} e_n + \sum_{k\geq n+1} \widetilde{e}_k e_{j-1}x^*\widetilde{e}_j x e_{j-1}\widetilde{e}_k \\ &= e_n x^*\widetilde{e}_j x e_n + \sum_{k=n+1}^{j-1} \widetilde{e}_k e_{j-1}x^*\widetilde{e}_j x e_{j-1}\widetilde{e}_k \\ &= e_n x^*\widetilde{e}_j x e_n + \sum_{k=n+1}^{j-1} \widetilde{e}_k x^*\widetilde{e}_j x \widetilde{e}_k,\end{aligned} \tag{3.16}$$

and

$$\mathcal{E}_n(\widetilde{e}_j x^* e_{j-1} x \widetilde{e}_j) = e_n[\widetilde{e}_j x^* e_{j-1} x \widetilde{e}_j]e_n + \sum_{k\geq n+1} \widetilde{e}_k[\widetilde{e}_j x^* e_{j-1} x \widetilde{e}_j]\widetilde{e}_k = \widetilde{e}_j x^* e_{j-1} x \widetilde{e}_j. \tag{3.17}$$

Combining (3.15), (3.16), and (3.17), we have

$$\begin{aligned}\sum_{j\geq n+1} \mathcal{E}_n(|d_j(x)|^2) &= \sum_{j\geq n+1} e_n x^*\widetilde{e}_j x e_n + \sum_{j\geq n+1}\sum_{k=n+1}^{j-1} \widetilde{e}_k x^*\widetilde{e}_j x \widetilde{e}_k + \sum_{j\geq n+1} \widetilde{e}_j x^* e_{j-1} x \widetilde{e}_j \\ &= e_n x^*(1-e_n)x e_n + \sum_{k\geq n+1} \widetilde{e}_k x^*(1-e_k)x\widetilde{e}_k + \sum_{j\geq n+1} \widetilde{e}_j x^* e_{j-1} x \widetilde{e}_j.\end{aligned}$$

This, together with (3.14) and (3.15) (with $j = n$ there) gives us the first desired equality. The second equality of the lemma follows from the first one. ■

Lemma 3.13. Let $\mathcal{Z}_{r,n}$ and $\mathcal{Z}_{c,n}$ be as in (2.3) and (2.4). For each $x \in \mathcal{BMO}^c(\mathcal{M})$, we have

$$\sup_n \|\mathcal{Z}_{c,n}(x)\|_\infty \leq 2\|x\|_{\mathcal{BMO}^c}.$$

Similarly,

$$\sup_n \|\mathcal{Z}_{r,n}(x)\|_\infty \leq 2\|x\|_{\mathcal{BMO}^r}.$$

Proof. We only show the first inequality since the second one can be proved similarly. For each $n \geq 2$, write

$$\|\mathcal{Z}_{c,n}(x)\|_\infty^2 = \||\mathcal{Z}_{c,n}(x)|^2\|_\infty = \|e_n x^*(1-e_{n-1})x e_n\|_\infty \tag{3.18}$$

and write

$$e_n x^*(1-e_{n-1})x e_n = e_n x^*(1-e_n)x e_n + e_n x^*\widetilde{e}_n x e_n.$$

By (3.2), we know that

$$e_n x^*\widetilde{e}_n x e_n \leq 2(e_{n-1}x^*\widetilde{e}_n x e_{n-1} + \widetilde{e}_n x^*\widetilde{e}_n x\widetilde{e}_n) \leq 2e_{n-1}x^*\widetilde{e}_n x e_{n-1} + 2\mathcal{E}_1(|x|^2).$$

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Hence, by Lemma 3.12, we deduce that

$$e_n x^*(1-e_{n-1})xe_n \le 2\mathcal{E}_n(|x-\mathcal{E}_{n-1}(x)|^2)+2\mathcal{E}_1(|x|^2). \tag{3.19}$$

Also note that

$$\|\mathcal{E}_1(|x|^2)\|_\infty = \|\mathcal{E}_1(|x-\mathcal{E}_0(x)|^2)\|_\infty \le \sup_n \|\mathcal{E}_n(|x-\mathcal{E}_{n-1}(x)|^2)\|_\infty = \|x\|^2_{\mathcal{BMO}^c}.$$

Therefore, combining (3.18) and (3.19), we obtain

$$\sup_n \|\mathcal{Z}_{c,n}(x)\|^2_\infty \le 2\sup_n \|\mathcal{E}_n(|x-\mathcal{E}_{n-1}(x)|^2)\|_\infty + 2\|\mathcal{E}_1(|x|^2)\|_\infty \le 4\|x\|^2_{\mathcal{BMO}^c}.$$

The first desired inequality follows. ■

Lemma 3.14. Let $\mathcal{Z}_{r,n}$ and $\mathcal{Z}_{c,n}$ be as in (2.3) and (2.4). For each $x \in L_2(\mathcal{M})$, we have

$$\|x\|_{\mathcal{BMO}} := \max\{\|x\|_{\mathcal{BMO}^c}, \|x\|_{\mathcal{BMO}^r}\} \le \sqrt{2}\max\{\sup_n \|\mathcal{Z}_{r,n}(x)\|_\infty, \sup_n \|\mathcal{Z}_{c,n}(x)\|_\infty\}.$$

Moreover, if $x \in BMO^c(\mathbb{A})$, then

$$\|x\|_{\mathcal{BMO}^c} \le \sqrt{2}\|x\|_{BMO^c(\mathbb{A})},$$

and if $x \in BMO^r(\mathbb{A})$, then

$$\|x\|_{\mathcal{BMO}^r} \le \sqrt{2}\|x\|_{BMO^r(\mathbb{A})}.$$

Proof. Denote by

$$Z_c := \sup_n \|\mathcal{Z}_{c,n}(x)\|_\infty = \sup_n \|e_n x^*(1-e_{n-1})xe_n\|^{1/2}_\infty$$

and

$$Z_r := \sup_n \|\mathcal{Z}_{r,n}(x)\|_\infty = \sup_n \|(1-e_{n-1})x^*e_n x(1-e_{n-1})\|^{1/2}_\infty.$$

Note that for each $j \ge 1$

$$\widetilde{e}_j x^* e_{j-1} x \widetilde{e}_j = \widetilde{e}_j(1-e_{j-1})x^*e_{j-1}x(1-e_{j-1})\widetilde{e}_j \le \widetilde{e}_j(1-e_{j-1})x^*e_j x(1-e_{j-1})\widetilde{e}_j \le \widetilde{e}_j Z_r^2$$

and

$$e_j x^*(1-e_j)xe_j = e_j e_{j+1}x^*(1-e_j)xe_{j+1}e_j \le e_j Z_c^2$$

Then it follows from Lemma 3.12 that for each $n \ge 1$,

$$\begin{aligned}
\mathcal{E}_n(|x-\mathcal{E}_{n-1}(x)|^2) &= e_{n-1}x^*\widetilde{e}_n x e_{n-1} + e_n x^*(1-e_n)xe_n + \sum_{k\ge n+1}\widetilde{e}_k x^*(1-e_k)x\widetilde{e}_k + \sum_{j\ge n}\widetilde{e}_j x^* e_{j-1}x\widetilde{e}_j \\
&\le e_{n-1}x^*(1-e_{n-1})xe_{n-1} + e_n Z_c^2 + \sum_{k\ge n+1}\widetilde{e}_k e_k x^*(1-e_k)xe_k\widetilde{e}_k + \sum_{k\ge n}\widetilde{e}_j Z_r^2 \\
&\le e_{n-1}Z_c^2 + e_n Z_c^2 + \sum_{k\ge n+1}\widetilde{e}_k Z_c^2 + \sum_{k\ge n}\widetilde{e}_j Z_r^2 \\
&\le 2Z_c^2 + (1-e_{n-1})Z_r^2.
\end{aligned}$$

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Consequently,

$$\|x\|_{\mathcal{BMO}^c} = \sup_n \|\mathcal{E}_n(|x - \mathcal{E}_{n-1}(x)|^2)\|_\infty^{1/2} \le \sqrt{2}\max\{Z_c, Z_r\}.$$

Similarly, we show

$$\|x\|_{\mathcal{BMO}^r} \le \sqrt{2}\max\{Z_c, Z_r\}.$$

The first inequality of the lemma follows.

As for $x \in BMO^c(\mathbb{A})$, we know that $e_k x = e_k x e_k$ for each $k \ge 1$. Repeating the above argument and using (3.13), we have

$$\|x\|_{\mathcal{BMO}^c} \le \sqrt{2}Z_c = \sqrt{2}\|x\|_{BMO^c(\mathbb{A})}.$$

Similarly, for $x \in BMO^r(\mathbb{A})$,

$$\|x\|_{\mathcal{BMO}^r} \le \sqrt{2}\|x\|_{BMO^r(\mathbb{A})}.$$

■

The space BMO^T introduced in [33] is given as follows:

$$BMO^T = \{x = y + Tz : y, z \in L_\infty(\mathcal{M})\},$$

equipped with the norm

$$\|x\|_{BMO^T} = \inf\{\|y\|_\infty + \|z\|_\infty\}.$$

The following lemma provides an equivalent description of the BMO^T-norm. It says that the duality holds:

$$(H_1^T(\mathcal{M}))^* = BMO^T,$$

which is known to experts (see, e.g., [33]). We write down the details for the reader's convenience.

Lemma 3.15. For each $x \in BMO^T$, we have

$$\frac{1}{3}\|x\|_{BMO^T} \le \sup\{|\tau(yx^*)| : \ \|T(y)\|_1, \|T^\perp(y)\|_1 \le 1, \ y \in L_1(\mathcal{M})\} \le 2\|x\|_{BMO^T}.$$

Proof. We first prove the right-hand side inequality. Take $x \in BMO^T$, and take $x_1, x_2 \in \mathcal{M}$ such that $x = x_1 + T(x_2)$ with

$$\|x_1\|_\infty + \|x_2\|_\infty \le \|x\|_{BMO^T}.$$

Let $y \in L_1(\mathcal{M})$ be such that $\|T(y)\|_1 \le 1$ and $\|T^\perp(y)\|_1 \le 1$. From this, we have $\|y\|_1 \le \|T(y)\|_1 + \|T^\perp(y)\|_1 \le 2$, and it is obvious that

$$\tau(yx^*) = \tau(yx_1^*) + \tau(y[T(x_2)]^*) = \tau(yx_1^*) + \tau(T(y)x_2^*).$$

Therefore,

$$|\tau(yx^*)| \le \|y\|_1\|x_1\|_\infty + \|T(y)\|_1\|x_2\|_\infty \le 2(\|x_1\|_\infty + \|x_2\|_\infty) \le 2\|x\|_{BMO^T}.$$

Taking the supremum over such y we obtain the right-hand side inequality.

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To prove the left-hand side inequality, denote the middle term by A. Let x be such that $|\tau(yx^*)| \le A$ for any $y \in L_1(\mathcal{M})$ with $\|T(y)\|_1 \le 1$ and $\|T^\perp(y)\|_1 \le 1$. In particular, we have

$$|\tau(yx^*)| \le A, \quad y = T(y),\ \|y\|_1 \le 1$$

and

$$|\tau(yx^*)| \le A, \quad y = T^\perp(y),\ \|y\|_1 \le 1.$$

By the Hahn-Banach theorem, there exist a functional f on $L_1(\mathcal{M})$ such that

$$|f(y)| \le A\|y\|_1, \quad y \in L_1(\mathcal{M})$$

and such that $f(y) = \tau(yx^*)$ for every $y = T(y)$. Similarly, there exist a functional g on $L_1(\mathcal{M})$ such that

$$|g(y)| \le A\|y\|_1, \quad y \in L_1(\mathcal{M})$$

and such that $g(y) = \tau(yx^*)$ for every $y = T^\perp(y)$. Since the Banach dual of $L_1(\mathcal{M})$ is $\mathcal{M}$, it follows that there exist $x_1, x_2 \in \mathcal{M}$ such that $\|x_1\|_\infty \le A$, $\|x_2\|_\infty \le A$ and such that

$$f(y) = \tau(yx_1^*), \quad g(y) = \tau(yx_2^*), \quad y \in L_1(\mathcal{M}).$$

Thus, $\tau(yx^*) = \tau(yx_1^*)$ for every $y = T(y)$ and $\tau(yx^*) = \tau(yx_2^*)$ for every $y = T^\perp(y)$. Hence, $T(x) = T(x_1)$ and $T^\perp(x) = T^\perp(x_2)$. It follows that

$$x = T(x) + T^\perp(x) = T(x_1) + T^\perp(x_2) = T(x_1 - x_2) + x_2, \quad \|x_1\|_\infty \le A,\ \|x_2\|_\infty \le A.$$

Consequently,

$$\|x\|_{BMO^T} \le \|x_1 - x_2\|_\infty + \|x_2\|_\infty \le 3A.$$

This proves the left-hand side inequality. ■

Theorem 3.16. The following are equivalent:

(i) $x \in \mathcal{BMO}(\mathcal{M}) := \mathcal{BMO}^r(\mathcal{M}) \cap \mathcal{BMO}^c(\mathcal{M})$;
(ii) $x \in BMO^T$; and
(iii) $\max\{\sup_n \|\mathcal{Z}_{r,n}(x)\|_\infty, \sup_n \|\mathcal{Z}_{c,n}(x)\|_\infty\} < \infty$.

Proof. Note that BMO^T is the dual of $H_1^T(\mathcal{M})$ (see its definition proceeding Lemma 3.9), and $\mathcal{BMO}(\mathcal{M}) = (\mathcal{H}_1^c(\mathcal{M}) + \mathcal{H}_1^r(\mathcal{M}))^*$ (see [43,p. 693]). Then, by Theorem 3.11, items (i) and (ii) are equivalent. By Lemma 3.13 and Lemma 3.14, we know that items (i) and (iii) are equivalent. The proof is complete. ■

Remark 3.17. A few comments are listed below.

(i) For the case $\mathcal{M} = B(H)$, Ovchinnikov [39, Theorem 2] previously announced that items (ii) and (iii) in last theorem are equivalent but not proved there. We later will provide a direct proof for this; see Corollary 5.8 below.
(ii) According to the noncommutative John-Nirenberg inequality established in [28, Theorem 1.1], the statement given in Theorem 3.16 is also equivalent to

$$\max\{\|x\|_{\mathcal{BMO}_p^c(\mathcal{M})}, \|x\|_{\mathcal{BMO}_p^r(\mathcal{M})}\} < \infty, \quad 2 < p < \infty,$$

where $\|x\|_{\mathcal{BMO}_p^r(\mathcal{M})} = \|x^*\|_{\mathcal{BMO}_p^c(\mathcal{M})}$ and

$$\|x\|_{\mathcal{BMO}_p^c(\mathcal{M})} = \sup_n \sup_{a \in L_p(\mathcal{M}_n), \|a\|_p \le 1} \|(x - \mathcal{E}_{n-1}(x))a\|_p.$$

(iii) Combining Theorem 3.16 and [28, Theorem 1.2], we have the following large deviation inequality for elements of BMO^T, that is, there exist universal constants $c_1, c_2 > 0$ such that if $x \in BMO^T$ with

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$\|x\|_{BMO^r} < 1$, then for any $t > 0$, there exists projection $e \in \mathcal{M}$ such that $\|(x - \mathcal{E}_1(x))e\|_{\mathcal{M}} \le t$ and

$$\tau(1-e) \le c_2 e^{-tc_1}.$$

The argument above also gives us an equivalent description of the $BMO^c(\mathbb{A})$-norm.

Theorem 3.18. Assuming that $x \in BMO^c(\mathbb{A})$, then

$$\|x\|_{BMO^c(\mathbb{A})} \approx \|x\|_{\mathcal{BMO}^c} \approx \|x\|_{\star,c},$$

where

$$\|x\|_{\star,c} = \sup_n \|e_n|x - e_n x e_n|^2 e_n\|_\infty^{1/2}.$$

Similar result holds for $x \in BMO^r(\mathbb{A})$.

Proof. Let $x \in BMO^c(\mathbb{A})$. By Lemma 3.13 and Lemma 3.14, we obtain

$$\|x\|_{BMO^c(\mathbb{A})} \approx \|x\|_{\mathcal{BMO}^c}.$$

Note also that for each $n \ge 1$

$$e_n|x - e_n x e_n|^2 e_n = e_n x^*(1-e_n)x e_n. \tag{3.20}$$

Thus, combining (3.20) and (3.13), we deduce that

$$\frac{1}{\sqrt{2}}\|x\|_{\mathcal{BMO}^c(\mathcal{M})} \le \sup_n \|e_n|x - e_n x e_n|^2 e_n\|_\infty^{1/2} \le \|x\|_{\mathcal{BMO}^c(\mathcal{M})}.$$

The proof is complete. ■

4 Noncommutative Maximal Inequalities

Let $\mathbb{A} = \{1, e_k\}_{k\ge 0}$ be as in Remark 2.3, and let

$$P_k(x) = e_k x e_k, \quad k \ge 1. \tag{4.1}$$

Theorem 4.2 proves noncommutative maximal inequality with respect to the family of operators $(P_k)_{k\ge1}$. We will see that Theorem 4.2 easily follows from noncommutative martingale Doob inequalities. However, we also would like to provide a new method which is based on triangular truncation operators. As a byproduct, we also obtain several new asymmetric Doob maximal inequalities.

Definition 4.1. Let $1 \le p \le \infty$. Assume that $A = (A_k)_{k\ge1}$ is a sequence of positive operators defined on $L_p(\mathcal{M})$.

(i) We say that A is of weak type (p,p) $(p < \infty)$ if for any $x \in L_p(\mathcal{M})$ and any $\lambda > 0$, there exist a constant $c > 0$ and a projection $e^\lambda \in \mathcal{M}$ such that

$$\tau(1-e^\lambda) \le \left(c\frac{\|x\|_p}{\lambda}\right)^p, \quad \text{and} \quad \sup_k \|e^\lambda A_k(x) e^\lambda\|_\infty \le \lambda.$$

(ii) We say that A is of type (p,p) if there exists a positive constant c_p such that

$$\|A(x)\|_{L_p(\mathcal{M},\ell_\infty)} \le c_p \|x\|_{L_p(\mathcal{M})}, \quad 1 \le p \le \infty.$$

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Theorem 4.2. Consider the sequence of positive operators $P = (P_k)_{k\geq 1}$ defined in (4.1). We have

(i) P is of weak type $(1,1)$.
(ii) P is of type (p,p), $1 < p \leq \infty$.

According to (2.12), for each $k \geq 1$ and positive $x \in \mathcal{M}$,

$$P_k(x) = e_k x e_k \leq \mathcal{E}_k(x).$$

Thus, with slight argument, Theorem 4.2 immediately follows from the following result which is taken from [11] and [25] (actually it holds for general noncommutative martingales).

Lemma 4.3. Consider the conditional expectations $\mathcal{E} = (\mathcal{E}_k)_{k\geq 1}$ given in Example 2.6. We have

(i) $\mathcal{E}$ is of weak type $(1,1)$.
(ii) $\mathcal{E}$ is of type (p,p), $1 < p \leq \infty$.

We are going to prove Theorem 4.2 via another method which is based on the triangular truncation operators. In what follows, for each $x \in \mathcal{M}$, write $\Delta_1(x) = P_1(x)$ and

$$\Delta_k(x) = P_k(x) - P_{k-1}(x), \quad k \geq 2.$$

Δ_k is known as the shell decomposition.

Lemma 4.4. Let $x \in \mathcal{M}$. For each $l \geq 1$, we have

(i) $\Delta_l(T(x)) = T(x)\widetilde{e}_l$;
(ii) $\Delta_l(T^{\perp}(x)) = \widetilde{e}_l T^{\perp}(x)$; and
(iii) $e_l x e_l = T(x)e_l + e_l T^{\perp}(x)$.

Proof. (i) Note that $\widetilde{e}_k e_l = e_l \widetilde{e}_k = 0$ provided $k \geq l+1$, and $\widetilde{e}_k e_l = e_l \widetilde{e}_k = \widetilde{e}_k$ for $k \leq l$. Hence, for each $l \geq 1$, we have

$$e_l T(x) e_l \overset{(3.1)}{=} e_l\Big(\sum_{k\geq 1}\sum_{j=1}^{k} \widetilde{e}_j x \widetilde{e}_k\Big)e_l = e_l\Big(\sum_{k=1}^{l}\sum_{j=1}^{k} \widetilde{e}_j x \widetilde{e}_k\Big)e_l$$
$$= \sum_{k=1}^{l}\sum_{j=1}^{k} \widetilde{e}_j x \widetilde{e}_k = \sum_{k=1}^{l} e_k x \widetilde{e}_k.$$

Since $(\widetilde{e}_k)_{k\geq 1}$ are pairwise disjoint, it follows that

$$T(x)\widetilde{e}_l = \Big(\sum_{k\geq 1}\sum_{j=1}^{k} \widetilde{e}_j x \widetilde{e}_k\Big)\widetilde{e}_l = \sum_{j=1}^{l} \widetilde{e}_j x \widetilde{e}_l = e_l x \widetilde{e}_l.$$

Hence,

$$e_l T(x) e_l - e_{l-1} T(x) e_{l-1} = \sum_{k=1}^{l} e_k x \widetilde{e}_k - \sum_{k=1}^{l-1} e_k x \widetilde{e}_k = e_l x \widetilde{e}_l = T(x)\widetilde{e}_l.$$

This shows item (i). Item (ii) can be proved similarly.

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For item (iii), combining the first two items, we obtain

$$\begin{aligned} e_l x e_l &= \sum_{i=1}^{l} \Delta_i(x) \\ &= \sum_{i=1}^{l} \Delta_i(T(x)) + \sum_{i=1}^{l} \Delta_i(T^{\perp}(x)) \\ &= \sum_{i=1}^{l} T(x)\widetilde{e}_l + \sum_{i=1}^{l} \widetilde{e}_l T^{\perp}(x) = T(x)e_l + e_l T^{\perp}(x). \end{aligned}$$

The proof is complete. ■

Proof of Theorem 4.2: based on triangular truncation operators. We only need to prove P is of weak type $(1,1)$. Indeed, since P is of type (∞,∞), once we show that P is weak type $(1,1)$, by the noncommutative Marcinkiewicz-type interpolation established in [29, Theorem 3.1], we obtain that P is of type (p,p) with $1 < p < \infty$.

Now we show that P is of weak type $(1,1)$. To this end, it suffices to show that for any given $\lambda > 0$, there exists a projection q^{λ} such that

$$\sup_k \|q^{\lambda} P_k(x) q^{\lambda}\|_{\infty} \le 2\lambda \tag{4.2}$$

and

$$\lambda\tau(1 - q^{\lambda}) \le c\|x\|_{L_1(\mathcal{M})}. \tag{4.3}$$

Set

$$q_1^{\lambda} = \chi_{[0,\lambda]}(|T(x)^*|), \quad q_2^{\lambda} = \chi_{[0,\lambda]}(|T^{\perp}(x)|)$$

and let $q^{\lambda} = q_1^{\lambda} \wedge q_2^{\lambda}$. Then, for each k, by Lemma 4.4 (iii), we have

$$\begin{aligned} \|q^{\lambda} P_k(x) q^{\lambda}\|_{\infty} &\le \|q^{\lambda} T(x) e_k q^{\lambda}\|_{\infty} + \|q^{\lambda} e_k T^{\perp}(x) q^{\lambda}\|_{\infty} \\ &\le \|q^{\lambda} T(x)\|_{\infty} + \|T^{\perp}(x) q^{\lambda}\|_{\infty} \\ &= \|q^{\lambda} T(x) T(x)^* q^{\lambda}\|_{\infty}^{1/2} + \|q^{\lambda} T^{\perp}(x)^* T^{\perp}(x) q^{\lambda}\|_{\infty}^{1/2}. \end{aligned}$$

Note that

$$\|q^{\lambda} T(x) T(x)^* q^{\lambda}\|_{\infty}^{1/2} = \|q^{\lambda} q_1^{\lambda} |T(x)^*|^2 q_1^{\lambda} q^{\lambda}\|_{\infty}^{1/2} \le \lambda.$$

Similarly, $\|q^{\lambda} T^{\perp}(x)^* T^{\perp} q^{\lambda}\|_{\infty}^{1/2} \le \lambda$. Therefore,

$$\|q^{\lambda} P_k(x) q^{\lambda}\|_{\infty} \le 2\lambda,$$

which proves (4.2). On the other hand, by Lemma 3.1, we know that

$$\begin{aligned} \lambda\tau(1 - q^{\lambda}) &\le \lambda\tau(1 - q_1^{\lambda}) + \lambda\tau(1 - q_2^{\lambda}) \\ &\le c\|x^*\|_{L_1(\mathcal{M})} + c\|x\|_{L_1(\mathcal{M})} = 2c\|x\|_{L_1(\mathcal{M})}, \end{aligned}$$

which shows (4.3). The proof is complete. ■

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Next we prove the dual to Theorem 4.2 (ii).

Corollary 4.5. For each finite positive sequence $(x_k)_{k\geq 1} \subset L_p(\mathcal{M})$, there exists $c_p > 0$ such that

$$\Big\| \sum_{k\geq 1} e_k x_k e_k \Big\|_{L_p(\mathcal{M})} \leq c_p \Big\| \sum_{k\geq 1} x_k \Big\|_{L_p(\mathcal{M})}, \quad 1 \leq p < \infty.$$

Proof. The case $p = 1$ is trivial. Here we only need to consider the case $1 < p < \infty$. By the duality $(L_p(\mathcal{M}))^* = L_{p'}(\mathcal{M})$, there exists positive $z \in L_{p'}(\mathcal{M})$ with norm 1 such that

$$\begin{aligned} \Big\| \sum_{k\geq 1} e_k x_k e_k \Big\|_{L_p(\mathcal{M})} &= \tau\Big(\sum_{k\geq 1} e_k x_k e_k z\Big) = \sum_{k\geq 1} \tau(x_k e_k z e_k) \\ &\overset{a}{\leq} \Big\| \sum_{k\geq 1} x_k \Big\|_{L_p(\mathcal{M})} \|(P_k(z))_k\|_{L_{p'}(\mathcal{M},\ell_\infty)} \\ &\overset{b}{\leq} c_p \Big\| \sum_{k\geq 1} x_k \Big\|_{L_p(\mathcal{M})}, \end{aligned}$$

where "a" is due to Lemma 2.2, and "b" is due to Theorem 4.2 (ii). ∎

We point out that Corollary 4.5 does not hold when $p = \infty$. If it did, by the duality result [25, Remark 3.7], it would imply that P is of type $(1, 1)$, and, by (2.12) and Lemma 2.7, $\mathcal{E}$ is of type $(1, 1)$. However, the latter contradicts to [27, Lemma 13].

Alternative proof of Corollary 4.5 is based on [25, Theorem 0.1]. Indeed, according to (2.12), for each $k \geq 1$ and positive x_k,

$$e_k x_k e_k \leq \mathcal{E}_k(x_k).$$

Hence,

$$\Big\| \sum_{k\geq 1} e_k x_k e_k \Big\|_{L_p(\mathcal{M})} \leq \Big\| \sum_{k\geq 1} \mathcal{E}_k(x_k) \Big\|_{L_p(\mathcal{M})} \leq c_p \Big\| \sum_{k\geq 1} x_k \Big\|_{L_p(\mathcal{M})},$$

where the last inequality is due to [25, Theorem 0.1].

Let us also discuss the asymmetric maximal inequalities which have independent interests. For any $x = (x_n)_{n\geq 1} \subset L_{1,\infty}(\mathcal{M})$, we define

$$\|x\|_{L_{1,\infty}(\mathcal{M},\ell_\infty^c)} = \inf\{\|a\|_{L_{1,\infty}} : a \geq 0, \quad |x_n|^2 \leq a^2, \forall n \geq 1\}.$$

Similarly, we define $\|x\|_{L_{1,\infty}(\mathcal{M},\ell_\infty^r)} = \|x^*\|_{L_{1,\infty}(\mathcal{M},\ell_\infty^c)}$. From Lemma 4.4 (iii) and Lemma 3.1, it is not hard to see that

$$\|\{P_n(Tx)\}_{n\geq 1}\|_{L_p(\mathcal{M},\ell_\infty^r)} + \|\{P_n(T^\perp x)\}_{n\geq 1}\|_{L_p(\mathcal{M},\ell_\infty^c)} \leq c_p \|x\|_p.$$

In spirit of this, we show the following result.

Proposition 4.6. Let $1 \leq p < \infty$.

(i) For $1 < p < \infty$ and $x \in L_p(\mathcal{M})$, there exists positive constant c_p such that

$$\|\{\mathcal{E}_n(Tx)\}_{n\geq 1}\|_{L_p(\mathcal{M},\ell_\infty^r)} + \|\{\mathcal{E}_n(T^\perp x)\}_{n\geq 1}\|_{L_p(\mathcal{M},\ell_\infty^c)} \leq c_p \|x\|_p,$$

where $c_p = O(1/(p-1))$ as $p \to 1$.

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(ii) For each $x \in L_1(\mathcal{M})$, there exists a positive constant c such that

$$\|\{\mathcal{E}_n(Tx)\}_{n\geq 1}\|_{L_{1,\infty}(\mathcal{M},\ell_\infty^r)} + \|\{\mathcal{E}_n(T^\perp x)\}_{n\geq 1}\|_{L_{1,\infty}(\mathcal{M},\ell_\infty^c)} \leq c\|x\|_1.$$

Proof. We only show item (i), item (ii) can be proved similarly. By symmetry and linearity, it suffices to show that

$$\|\{\mathcal{E}_n(T^1 x)\}_{n\geq 1}\|_{L_p(\mathcal{M},\ell_\infty^r)} \leq c_p\|x\|_p,$$

where $T^1 x = Tx - \mathcal{E}_1(x)$. According to Proposition 3.4 and Lemma 3.3, we know that

$$T^1 x = \sum_{k\geq 2} e_{k-1} d_k(x) = \sum_{k\geq 2} e_{k-1} x \widetilde{e}_k.$$

By (2.12), we have

$$\mathcal{E}_n(T^1 x) = (T_1 x)e_n, \quad \forall n \geq 1.$$

Hence, $|\mathcal{E}_n(T^1 x)^*|^2 \leq |(T^1 x)^*|^2$, and consequently,

$$\|\{\mathcal{E}_n(T^1 x)\}_{n\geq 1}\|_{L_p(\mathcal{M},\ell_\infty^r)} \leq \|T^1 x\|_p \leq \|Tx\|_p + \|\mathcal{E}_1(x)\|_p \leq c_p\|x\|_p,$$

where we used Lemma 3.1 and Lemma 2.7 in the last inequality. The behavior of the constant c_p is because of Lemma 3.1.

To prove item (ii), with the same argument, we need to apply the weak estimates for triangular truncation operators in Lemma 3.1. The proof is complete. ■

Remark 4.7.

(i) Let $1 < p < 2$. It was proved in [17, Theorem B] that for general noncommutative martingales, there exists a positive constant K_p such that

$$\inf_{x=y+z}\{\|\{\mathcal{E}_n(y)\}_{n\geq 1}\|_{L_p(\mathcal{M},\ell_\infty^r)} + \|\{\mathcal{E}_n(z)\}_{n\geq 1}\|_{L_p(\mathcal{M},\ell_\infty^c)}\} \leq K_p\|x\|_p, \quad x \in L_p(\mathcal{M}). \tag{4.4}$$

In spirit of Proposition 4.6 (i), it is natural to ask whether one can find a concrete decomposition $x = y + z$ such that the last inequality holds.

(ii) Note that Proposition 4.6 (ii) can be viewed as endpoint case of (4.4). We do not know whether Proposition 4.6 (ii) holds for general noncommutative martingales.

We conclude this section with the following result, which is comparable with the noncommutative Stein inequality established in [43, Theorem 2.3].

Proposition 4.8. Let $1 < p < \infty$. For any finite sequence $(x_k)_{k\geq 1} \subset L_p(\mathcal{M})$, we have

$$\Big\|\Big(\sum_{k\geq 1}|x_k e_k|^2\Big)^{1/2}\Big\|_{L_p(\mathcal{M})} \leq c_p\Big\|\Big(\sum_{k\geq 1}|x_k|^2\Big)^{1/2}\Big\|_{L_p(\mathcal{M})}.$$

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Proof. We first show the case $2 \le p < \infty$. By Corollary 4.5, we have

$$\Big\|\Big(\sum_{k\ge1}|x_k e_k|^2\Big)^{1/2}\Big\|_{L_p(\mathcal{M})} = \Big\|\sum_{k\ge1} e_k|x_k|^2 e_k\Big\|^{1/2}_{L_{p/2}(\mathcal{M})}$$
$$\le c_p\Big\|\sum_{k\ge1}|x_k|^2\Big\|^{1/2}_{L_{p/2}(\mathcal{M})} = c_p\Big\|\Big(\sum_{k\ge1}|x_k|^2\Big)^{1/2}\Big\|_{L_p(\mathcal{M})}.$$

Now we consider the case $1 < p < 2$. By the duality (2.8), there exists a sequence of operators in $L_{p'}(\mathcal{M})$ with $\|(\sum_{k\ge1}|y_k|^2)^{1/2}\|_{L_{p'}(\mathcal{M})} \le 1$ such that

$$\Big\|\Big(\sum_{k\ge1}|x_k e_k|^2\Big)^{1/2}\Big\|_{L_p(\mathcal{M})} = \Big|\sum_{k\ge1}\tau(x_k e_k y_k^*)\Big|$$
$$\le \Big\|\Big(\sum_{k\ge1}|x_k|^2\Big)^{1/2}\Big\|_{L_p(\mathcal{M})}\Big\|\Big(\sum_{k\ge1}|y_k e_k|^2\Big)^{1/2}\Big\|_{L_p(\mathcal{M})}$$
$$\le c_p\Big\|\Big(\sum_{k\ge1}|x_k|^2)^{1/2}\Big\|_{L_p(\mathcal{M})}\Big\|\Big(\sum_{k\ge1}|y_k|^2\Big)^{1/2}\Big\|_{L_{p'}(\mathcal{M})}$$
$$\le c_p\Big\|\Big(\sum_{k\ge1}|x_k|^2\Big)^{1/2}\Big\|_{L_p(\mathcal{M})},$$

where the first inequality is due to Lemma 2.8, and the second inequality is due to the assertion we obtained in last paragraph. ■

5 Atomic Decomposition, Duality, and Interpolation

In this section, we first establish atomic decomposition of the Hardy space $H_p^r(\mathbb{A})$. With the help of the atomic decomposition, we study the dual space of Hardy space $H_p^r(\mathbb{A})$ with $1 \le p < 2$, and then we study complex interpolation between $H_p^r(\mathbb{A})$ and $BMO^r(\mathbb{A})$ spaces. Here the symbol $\mathbb{A}$ is explained in Remark 2.3.

5.1 Atomic decomposition

In this subsection, we aim to show the following atomic decomposition via the Riesz factorization Theorem 2.4.

Theorem 5.1. Let $0 < p, q, s \le \infty$ with $1/s = 1/p + 1/q$. Then, $x \in H_s^r(\mathbb{A})$ if and only if there exist $(a_k)_{k\ge1} \subset L_p(\mathcal{M})$ and $(b_k)_{k\ge1} \subset L_q(\mathcal{M})$ such that

(i) $x = \sum_{k\ge1} a_k b_k$ in $L_s(\mathcal{M})$;
(ii) $(1 - e_k)a_k = b_k e_{k-1} = 0$, $k \ge 1$;

and

$$\|x\|_s = \inf\Big\|\Big(\sum_{k\ge1}|a_k^*|^2\Big)^{1/2}\Big\|_p\Big\|\Big(\sum_{k\ge1}|b_k|^2\Big)^{1/2}\Big\|_q, \tag{5.1}$$

where the infimum is taken over all decompositions satisfying (i) and (ii).

Proof. Take $x \in H_s^r(\mathbb{A})$. According to Theorem 2.4, for any $\epsilon > 0$, there exist $y \in H_p^r(\mathbb{A})$ and $z \in H_q^r(\mathbb{A})$ such that $x = yz$ and

$$\|x\|_s \le \|y\|_p\|z\|_q < \|x\|_r + \epsilon. \tag{5.2}$$

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Recall that $(\widetilde{e}_k)_{k\geq 1}$ is defined in (2.10). For each k, set $a_k = y\widetilde{e}_k$ and $b_k = \widetilde{e}_k z$. Then,

$$x = yz = \sum_{k\geq 1} y\widetilde{e}_k z = \sum_{k\geq 1} a_k b_k.$$

It is not hard to see that $\sum_{k\geq 1} a_k b_k$ converges in $L_s(\mathcal{M})$. The item (i) is verified. Note that, for each k, we have (note that $y \in H_p^r(\mathbb{A})$)

$$(1-e_k)a_k = (1-e_k)y\widetilde{e}_k = (1-e_k)e_k y e_k \widetilde{e}_k = 0.$$

Similarly, for each k, we have

$$b_k e_{k-1} = \widetilde{e}_k z e_{k-1} = \widetilde{e}_k e_{k-1} z e_{k-1} = 0.$$

Then item (ii) is verified. Finally, let us verify (5.1). Observe that

$$\begin{aligned}\|y\|_p &= \|(yy^*)^{1/2}\|_p \\ &= \Big\|\Big(\sum_{k=1}^m y\widetilde{e}_k y^*\Big)^{1/2}\Big\|_p = \Big\|\Big(\sum_{k=1}^m a_k a_k^*\Big)^{1/2}\Big\|_p\end{aligned}$$

and, by analogy,

$$\|z\|_q = \Big\|\Big(\sum_{k=1}^m |b_k|^2\Big)^{1/2}\Big\|_q.$$

Now (5.1) follows from (5.2).

Now we show the converse part. Let $x = \sum_{k\geq 1} a_k b_k \in L_s(\mathcal{M})$ with $(a_k)_{k\geq 1}$ and $(b_k)_{k\geq 1}$ satisfying (ii). By the Hölder inequality Lemma 2.8, (5.1) holds. Hence, to show $x \in H_s^r(\mathbb{A})$, it suffices to show that

$$xe_j = e_j x e_j, \quad j \geq 1.$$

Indeed, for each j, we have

$$\begin{aligned}xe_j &= \sum_{k\geq 1} a_k b_k e_j \\ &= \sum_{k=1}^j a_k b_k e_j + \sum_{k\geq j+1} a_k b_k e_j \overset{\text{(ii)}}{=} \sum_{k=1}^j a_k b_k e_j + 0 = \sum_{k=1}^j a_k b_k e_j.\end{aligned}$$

On the other hand, by (ii), we know that, for each $k \geq 1$,

$$a_k = (1-e_k)a_k + e_k a_k = e_k a_k.$$

Then, for each j,

$$e_j x e_j = e_j\Big(\sum_{k=1}^j a_k b_k e_j\Big) = e_j\Big(\sum_{k=1}^j e_k a_k b_k e_j\Big) = \sum_{k=1}^j a_k b_k e_j = xe_j.$$

The desired assertion is proved, and hence, the proof of the theorem is complete. ■

Remark 5.2. The above result is comparable with the algebraic atomic decomposition of noncommutative martingale Hardy spaces studied in [9, Theorem 3.10] (see also [17], [50] and [23]).

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5.2 Duality

It was proved in [26, Theorem 4.1] for noncommutative martingale Hardy spaces that for $1 \le p < 2$,

$$(\mathcal{H}_p^c(\mathcal{M}))^* = \mathcal{L}_q^c\mathcal{MO}(\mathcal{M}), \frac{1}{p} + \frac{1}{q} = 1,$$

with equivalent norms. In spirit of this, we show the following duality. The space $L_q^c MO(\mathbb{A})$ below is given in (2.5).

Theorem 5.3. Let $1 \le p < 2$ and $1/p + 1/q = 1$. Then

$$(H_p^r(\mathbb{A}))^* = L_q^r MO(\mathbb{A})$$

with equivalent norms. The same result holds for $H_p^c(\mathbb{A})$ and $L_q^c MO(\mathbb{A})$.

To show this, we need the following lemma. The notation $\mathcal{Z}_{r,k}$ is referred to (2.3).

Lemma 5.4. Let $1 \le p < 2$ and q be the conjugate index of p. For each $y \in L_q(\mathcal{M})$ and $x \in H_p^r(\mathbb{A})$ with $\|x\|_p \le 1$, we have

$$|\tau(xy^*)| \le \|(\mathcal{Z}_{r,k}(y))_k\|_{L_q(\mathcal{M},\ell_\infty^r)}.$$

Similarly, for each $y \in L_q(\mathcal{M})$ and $x \in H_p^c(\mathbb{A})$ with $\|x\|_p \le 1$, we have

$$|\tau(xy^*)| \le \|(\mathcal{Z}_{r,k}(y))_k\|_{L_q(\mathcal{M},\ell_\infty^c)}.$$

Proof. For $y \in L_q(\mathcal{M})$, $\|(\mathcal{Z}_{r,k}(y))_k\|_{L_q(\mathcal{M},\ell_\infty^r)} < \infty$ by Lemma 5.12. According to Theorem 5.1, for $x \in H_p^r(\mathbb{A})$, there exist $(a_k)_{k\ge1} \subset L_s(\mathcal{M})$ and $(b_k)_{k\ge1} \subset L_2(\mathcal{M})$ such that (the number s is chosen to satisfy $1/p = 1/2 + 1/s$)

(i) $x = \sum_{k\ge1} a_k b_k$;
(ii) $(1 - e_k)a_k = b_k e_{k-1} = 0$, $k \ge 1$; and
(iii) the estimate holds:

$$\left\|\left(\sum_{k\ge1} |a_k^*|^2\right)^{1/2}\right\|_s, \quad \left\|\left(\sum_{k\ge1} |b_k|^2\right)^{1/2}\right\|_2 \le 1.$$

Also, observe that by (ii),

$$\mathcal{Z}_{r,k}(a_k b_k) = e_k a_k b_k (1 - e_{k-1}) = a_k b_k, \quad k \ge 1.$$

Thus,

$$\begin{aligned} |\tau(y^*x)| &= \left|\sum_{k\ge1} \tau(y^* a_k b_k)\right| \\ &= \left|\sum_{k\ge1} \tau(y^* \mathcal{Z}_{r,k}(a_k b_k))\right| = \left|\sum_{k\ge1} \tau(\mathcal{Z}_{r,k}(y)^* a_k b_k)\right| \\ &\le \left(\sum_{k\ge1} \|\mathcal{Z}_{r,k}(y)^* a_k\|_2^2\right)^{1/2} \left(\sum_{k\ge1} \|b_k\|_2^2\right)^{1/2} \\ &\le \|(\mathcal{Z}_{r,k}(y))_k\|_{L_q(\mathcal{M},\ell_\infty^r)}, \end{aligned} \tag{5.3}$$

where the last inequality is due to Lemma 2.1 since $1/q + 1/s = 1/2$. The second inequality of the lemma can be similarly proved. ∎

Now we prove Theorem 5.3.

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Proof of Theorem 5.3. Let $1 \le p < 2$ and $1/p + 1/q = 1$. We first show $L_q^r MO(\mathbb{A}) \subset (H_p^r(\mathbb{A}))^*$. For each $y \in L_q^r MO(\mathbb{A})$, define a linear functional on $H_p^r(\mathbb{A})$ by

$$\ell_y(x) = \tau(xy^*), \quad x \in H_p^r(\mathbb{A}).$$

Without loss of generality, assume that $\|x\|_p \le 1$. It follows from Lemma 5.4 that

$$\|\ell_y\| = \sup_{x \in H_p^r(\mathbb{A}), \|x\|_p \le 1} |\tau(y^*x)| \le \|(\mathcal{Z}_{r,k}(y))_k\|_{L_q(\mathcal{M}, \ell_\infty^r)} = \|y\|_{L_q^r MO},$$

which says $L_q^r MO(\mathbb{A}) \subset (H_p^r(\mathbb{A}))^*$.

Now we turn to the converse part. Let $\ell \in (H_p^r(\mathbb{A}))^*$ be such that $\|\ell\|_{(H_p^r(\mathbb{A}))^*} \le 1$. Since $H_2^r(\mathbb{A}) \subset H_p^r(\mathbb{A})$, it follows that ℓ induces a continuous functional $\widetilde{\ell}$ on $H_2^r(\mathbb{A})$. Hence, there exists $y \in H_2^r(\mathbb{A})$ such that

$$\widetilde{\ell}(x) = \tau(y^*x), \quad x \in H_2(\mathbb{A}).$$

By the density of $H_2^r(\mathbb{A})$ in $H_p^r(\mathbb{A})$, we have

$$\|\ell\|_{(H_p^r(\mathbb{A}))^*} = \sup_{x \in H_2^r(\mathbb{A}), \|x\|_p \le 1} |\tau(y^*x)| \le 1. \tag{5.4}$$

We need to show

$$\|y\|_{L_q^r MO} = \|(\mathcal{Z}_{r,k}(y))_k\|_{L_q(\mathcal{M}, \ell_\infty^r)} = \|(|\mathcal{Z}_{r,k}(y)^*|^2)_k\|_{L_{\frac{q}{2}}(\mathcal{M}, \ell_\infty)}^{1/2} < \infty.$$

Let α be the conjugate index of $q/2$, and it is easy to see that $\alpha = s/2$ (s is chosen to satisfy $1/p = 1/2 + 1/s$). By Lemma 2.2, there exists a sequence of positive operators $(b_k)_k \subset L_\alpha(\mathcal{M})$ such that $\|\sum_k b_k\|_{L_\alpha(\mathcal{M})} \le 1$ and

$$\begin{aligned}
\|y\|_{L_q^r MO}^2 &= \sum_{k \ge 1} \tau(|\mathcal{Z}_{r,k}(y)^*|^2 b_k) \\
&= \sum_{k \ge 1} \tau([e_k y(1 - e_{k-1}) y^* e_k] b_k) \\
&= \sum_{k \ge 1} \tau(y^*[e_k b_k e_k y(1 - e_{k-1})]),
\end{aligned}$$

where the first equality is due to the definition of $\mathcal{Z}_{r,k}$ given in (2.3), and the second equality is due to the tracial property. Define

$$x = \sum_{k \ge 1} e_k b_k e_k y(1 - e_{k-1}) = \sum_{k \ge 1} e_k b_k^{1/2} \cdot b_k^{1/2} e_k y(1 - e_{k-1}).$$

By Theorem 5.1, we have that $x \in H_p^r(\mathbb{A})$, and moreover

$$\begin{aligned}
\|x\|_p &\le \Big\| \Big(\sum_{k \ge 1} |(e_k b_k^{1/2})^*|^2 \Big)^{1/2} \Big\|_s \Big\| \Big(\sum_{k \ge 1} |b_k^{1/2} e_k y(1 - e_{k-1})|^2 \Big)^{1/2} \Big\|_2 \\
&\le c_{s_1} \Big\| \Big(\sum_{k \ge 1} |b_k^{1/2}|^2 \Big)^{1/2} \Big\|_s \Big[\sum_{k \ge 1} \tau([e_k y(1 - e_{k-1}) y^* e_k] b_k) \Big]^{1/2} \\
&= c_s \Big\| \sum_{k \ge 1} b_k \Big\|_{\frac{s}{2}}^{1/2} [\tau(y^*x)]^{1/2} = c_s \Big\| \sum_{k \ge 1} b_k \Big\|_\alpha^{1/2} [\tau(y^*x)]^{1/2} \\
&\le c_s [\tau(y^*x)]^{1/2},
\end{aligned}$$

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where the second inequality is due to Proposition 4.8. Hence, combining (5.4) and the proceeding inequality, we have

$$\|y\|^2_{L^r_q MO} = \tau(y^*x) \le \|x\|_p \le c_{s_1}[\tau(y^*x)]^{1/2},$$

which implies that $y \in L^r_q MO(\mathbb{A})$ with

$$\|y\|_{L^r_q MO} \le \sqrt{c_s}.$$

The proof is complete. ∎

Denote

$$BMO^{T,r} = BMO^T \cap H^r_2(\mathbb{A}),$$

where the definition of BMO^T is given proceeding Theorem 3.16. The following results are special case of Marsalli and West's results proved in [33, Theorem 5] and [34, Corollary 6.3]:

$$(H^r_1(\mathbb{A}))^* = BMO^{T,r},$$

and

$$(H^r_p(\mathbb{A}))^* = H^r_q(\mathbb{A}), \quad 1 < p < \infty, \frac{1}{p} + \frac{1}{q} = 1. \tag{5.5}$$

Note that $L^r_\infty MO(\mathbb{A})$ is denoted by $BMO^r(\mathbb{A})$. By Marsalli and West's results above, the following corollaries follow from Theorem 5.3.

Corollary 5.5. We have

$$BMO^r(\mathbb{A}) = BMO^{T,r}$$

with equivalent norms.

Corollary 5.6. For $2 < q < \infty$, we have

$$H^r_q(\mathbb{A}) = L^r_q MO(\mathbb{A})$$

with equivalent norms. The same result holds for $H^c_q(\mathbb{A})$ and $L^c_q MO(\mathbb{A})$.

The following result shows that every upper triangular matrix in BMO space has a bounded extension. It goes back to [39, Theorem 1] when $\mathcal{M} = B(H)$ is the space of all bounded linear operators on some Hilbert space H.

Corollary 5.7. Let $x \in H^r_\infty(\mathbb{A})$. The following assertions are equivalent:

(i) x has a bounded extension, that is, there exists $\widetilde{x} \in \mathcal{M}$ such that $T(\widetilde{x}) = x$.
(ii) We have

$$\|x\|_{BMO^r(\mathbb{A})} = \sup_k \|\mathcal{Z}_{r,k}(x)\|_\infty < \infty.$$

Proof. Assume item (i) holds. It follows from Corollary 3.6 that

$$\|x\|_{\mathcal{BMO}^r} = \|T(\widetilde{x})\|_{\mathcal{BMO}^r} \le 2\|\widetilde{x}\|_{\mathcal{BMO}^r} \le 4\|\widetilde{x}\|_{\mathcal{M}} < \infty,$$

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where in the last inequality we used the fact $\|y\|_{\mathcal{BMO}^r} \leq 2\|y\|_{\mathcal{M}}$ for each $y \in \mathcal{M}$ (see, e.g., [43,p.693]). Now item (ii) follows from Lemma 3.13.

Conversely, according to Theorem 5.3 with $p = 1$ there, for each $x \in BMO^r(\mathbb{A})$, x can be viewed as a continuous linear functional on $H_1^r(\mathbb{A})$. Also note that $H_1^r(\mathbb{A})$ is a closed subspace of $L_1(\mathcal{M})$. Hence, by the Hahn–Banach extension theorem, x has an extension, $\widetilde{x}$, which is a continuous linear functional on $L_1(\mathcal{M})$, that is, $\widetilde{x} \in \mathcal{M}$. Moreover,

$$\tau(\widetilde{x}^* y) = \tau(x^* y), \quad \forall y \in H_1^r(\mathbb{A}).$$

Since $y = T(y)$ for each $y \in H_1^r(\mathbb{A})$ and T is self-adjoint, it follows that

$$\tau(\widetilde{x}^* y) = \tau(\widetilde{x}^* T(y)) = \tau([T(\widetilde{x})]^* y) = \tau(x^* y), \quad \forall y \in H_1^r(\mathbb{A}).$$

Therefore, $T(\widetilde{x}) = x$. ■

We would also like to provide another proof to show the result below. It goes back to [39, Theorem 2] when $\mathcal{M} = B(H)$.

Corollary 5.8. Items (ii) and (iii) of Theorem 3.16 are equivalent.

Proof. We start with (ii) ⇒ (iii). Take $x \in BMO^T$. According to the definition of BMO^T, we can find $y, z \in \mathcal{M}$ such that $x = y + T(z)$ and

$$\|y\|_\infty + \|z\|_\infty \leq \|x\|_{BMO^T}.$$

Basic calculation gives us that for each $k \geq 1$,

$$\mathcal{Z}_{c,k}(T(z)) = (1 - e_{k-1})T(z)e_k = \widetilde{e}_k z \widetilde{e}_k,$$

and

$$\mathcal{Z}_{r,k}(T(z)) = e_k T(z)(1 - e_{k-1}) = e_k z(1 - e_{k-1}).$$

Thus,

$$\begin{aligned}\sup_k \|\mathcal{Z}_{c,k}(x)\|_\infty &\leq \sup_k \|\mathcal{Z}_{c,k}(y)\|_\infty + \sup_k \|\mathcal{Z}_{c,k}(T(z))\|_\infty \\ &= \sup_k \|(1 - e_{k-1})ye_k\|_\infty + \sup_k \|\widetilde{e}_k z \widetilde{e}_k\|_\infty \\ &\leq \|y\|_\infty + \|z\|_\infty \leq \|x\|_{BMO^T}.\end{aligned}$$

Similarly, we have $\sup_k \|\mathcal{Z}_{r,k}(x)\|_\infty \leq \|x\|_{BMO^T}$, and consequently,

$$\max\{\sup_k \|\mathcal{Z}_{r,k}(x)\|_\infty, \sup_k \|\mathcal{Z}_{c,k}(x)\|_\infty\} \leq \|x\|_{BMO^T},$$

which is just (iii) of Theorem 3.16.

Now we show (iii) ⇒ (ii). Take $x \in L_2(\mathcal{M})$ such that $\max\{\sup_n \|\mathcal{Z}_{r,n}(x)\|_\infty, \sup_n \|\mathcal{Z}_{c,n}(x)\|_\infty\} < \infty$. Take $y \in \mathcal{M}$ such that $\|T(y)\|_1 \leq 1$ and $\|T^\perp(y)\|_1 \leq 1$. This means $T(y) \in H_1^r(\mathbb{A})$ and $T^\perp(y) \in H_1^c(\mathbb{A})$ with norm

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less 1. According to Lemma 5.4, we deduce that

$$\begin{aligned}|\tau(yx^*)| &= |\tau(T(y)x^*) + \tau(T^\perp(y)x^*)|\\ &\le |\tau(T(y)x^*)| + |\tau(T^\perp(y)x^*)|\\ &\le \|(\mathcal{Z}_{r,k}(x))_k\|_{L_q(\mathcal{M},\ell_\infty^r)} + \|(\mathcal{Z}_{c,k}(x))_k\|_{L_q(\mathcal{M},\ell_\infty^c)}\\ &\le 2\max\{\sup_k \|\mathcal{Z}_{r,k}(x)\|_\infty, \sup_k \|\mathcal{Z}_{c,k}(x)\|_\infty\}.\end{aligned}$$

Consequently, by Lemma 3.15, we have

$$\begin{aligned}\|x\|_{BMO^T} &\le 3\sup\{|\tau(yx^*)| : \ \|T(y)\|_1, \|T^\perp(y)\|_1 \le 1, \ y \in \mathcal{M}\}\\ &\le 6\max\{\sup_n \|\mathcal{Z}_{r,n}(x)\|_\infty, \sup_n \|\mathcal{Z}_{c,n}(x)\|_\infty\}.\end{aligned}$$

The proof is complete. ■

5.3 Complex interpolation

Let (X_0, X_1) be an interpolation couple of Banach spaces; namely $X_j, j = 0, 1$ are continuously embedded into a Hausdorff topological vector space Y. Let $\mathcal{S}$ (respectively, $\overline{\mathcal{S}}$) denote the open strip $\{z : 0 < \mathrm{Re}z < 1\}$ (respectively, the closed strip $\{z : 0 \le \mathrm{Re}z \le 1\}$) in the complex plane $\mathbb{C}$. Denote by $\partial_0 = \{z \in \overline{\mathcal{S}} : \mathrm{Re}z = 0\}$, $\partial_1 = \{z \in \overline{\mathcal{S}} : \mathrm{Re}z = 1\}$ the boundaries of $\overline{\mathcal{S}}$. Let $\mathcal{F}(X_0, X_1)$ be the space of bounded analytic functions $f : \mathcal{S} \to X_0 + X_1$ which extend continuously to $\overline{\mathcal{S}}$ such that the functions $t \mapsto f(j + it)$ are bounded and continuous from $\mathbb{R}$ into $X_j, j = 0, 1$. We equip $\mathcal{F}(X_0, X_1)$ with the norm

$$\|f\|_{\mathcal{F}(X_0,X_1)} = \max\left\{\sup_{z\in\partial_0}\|f(z)\|_{X_0}, \sup_{z\in\partial_1}\|f(z)\|_{X_1}\right\}.$$

Then $\mathcal{F}(X_0, X_1)$ is a Banach space. For $0 \le \theta \le 1$, we define the complex interpolation space $[X_0, X_1]_\theta$ as the set of all $x \in X_0 + X_1$ satisfying that $x = f(\theta)$ for some $f \in \mathcal{F}(X_0, X_1)$. The norm on $[X_0, X_1]_\theta$ is defined by setting

$$\|x\|_{[X_0,X_1]_\theta} = \inf\left\{\|f\|_{\mathcal{F}(X_0,X_1)} : f \in \mathcal{F}(X_0, X_1), f(\theta) = x\right\}.$$

It then follows that $[X_0, X_1]_\theta$ is a Banach space for $0 \le \theta \le 1$ (see [5, Theorem 4.1.2]).

Complex interpolation between $H_p^r(\mathbb{A})$ and $H_\infty^r(\mathbb{A})$ was first studied by Pisier [42]. We here concern the complex interpolation between $H_p^r(\mathbb{A})$ and $BMO^r(\mathbb{A})$ (which contains $H_\infty^r(\mathbb{A})$). Recall that Musat [38] proved in the noncommutative martingale setting that

$$[\mathcal{BMO}^r(\mathcal{M}), \mathcal{H}_p^r(\mathcal{M})]_{\frac{p}{q}} = \mathcal{H}_q^r(\mathcal{M}).$$

We here provide a proof to the following result by the techniques imported from [38].

Theorem 5.9. Let $1 \le p < q < \infty$. We have

$$[BMO^r(\mathbb{A}), H_p^r(\mathbb{A})]_{\frac{p}{q}} = H_q^r(\mathbb{A}).$$

The same result holds for column spaces.

We need the following proposition whose proof is provided after the proof of Theorem 5.9.

Proposition 5.10. Let $2 \le p < q < \infty$. Then,

$$[BMO^r(\mathbb{A}), L_p^r MO(\mathbb{A})]_{\frac{p}{q}} = L_q^r MO(\mathbb{A}).$$

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Combining Corollary 5.6 and the last proposition, we obtain

$$[BMO^r(\mathbb{A}), H_p^r(\mathbb{A})]_{\frac{p}{q}} = H_q^r(\mathbb{A}), \quad 2 < p < q < \infty. \tag{5.6}$$

We will continue to finish the proof of Theorem 5.9 via Wolff's interpolation theorem [54, Theorem 2] and duality argument.

The Wolff's interpolation theorem states that for Banach spaces $\{A_i\}_{i=1}^4$, if Banach space $A_1 \cap A_4$ is dense in both A_2 and A_3, $A_2 = [A_1, A_3]_\theta$ and $A_3 = [A_2, A_4]_\phi$ provided $0 < \theta, \phi < 1$, then

$$A_2 = [A_1, A_4]_\zeta, \quad A_3 = [A_1, A_4]_\xi, \tag{5.7}$$

where $\zeta = \frac{\theta\phi}{1-\theta+\theta\phi}$ and $\xi = \frac{\phi}{1-\theta+\theta\phi}$.

Proof of Theorem 5.9. We prove the theorem by considering cases: $1 < p < q < \infty$ and $1 = p < q < \infty$.

Case 1: $1 < p < q < \infty$. We first assume that $2 < q < \infty$. Note that for $1 < a < \infty$, $H_a^r(\mathbb{A})$ is complemented in $L_a(\mathcal{M})$ (see, e.g., Lemma 3.1), and the complex interpolation between noncommutative Lebesgue spaces $L_p(\mathcal{M})$ is well known. There exists $s \in (q, \infty)$ such that

$$A_3 := H_q^r(\mathbb{A}) = [H_s^r(\mathbb{A}), H_p^r(\mathbb{A})]_\phi = [A_2, A_4]_\phi, \quad \frac{1-\phi}{s} + \frac{\phi}{p} = \frac{1}{q}.$$

By (5.6), we get

$$A_2 := H_s^r(\mathbb{A}) = [BMO^r(\mathbb{A}), H_q^r(\mathbb{A})]_\theta = [A_1, A_3]_\theta, \quad \frac{\theta}{q} = \frac{1}{s}.$$

Thus, using (5.7), we obtain

$$A_3 = H_q^r(\mathbb{A}) = [A_1, A_4]_\xi = [BMO^r(\mathbb{A}), H_p^r(\mathbb{A})]_\xi,$$

where $\xi = \frac{\phi}{1-\theta+\theta\phi} = \frac{p}{q}$. This, together with (5.7), further implies desired result for the range $1 < p < q < \infty$.

Case 2: $1 = p < q < \infty$. Take $s > \max\{q, 2\}$. Then there exists $\phi \in (0, 1)$ such that $\frac{1-\phi}{s} + \phi = \frac{1}{q}$. On the other hand, combining [5, Corollary 4.5.2], the duality results (see Theorem 5.3 and (5.5)), and Case 1 that

$$([H_s^r(\mathbb{A}), H_1^r(\mathbb{A})]_\phi)^* = [(H_s^r(\mathbb{A}))^*, (H_1^r(\mathbb{A}))^*]_\phi = [H_{s'}^r(\mathbb{A}), BMO^r(\mathbb{A})]_\phi = H_{q'}^r(\mathbb{A}).$$

Thus,

$$A_3 = H_q^r(\mathbb{A}) = [H_s^r(\mathbb{A}), H_1^r(\mathbb{A})]_\phi = [A_2, A_4]_\phi.$$

Note that $s > 2$. By Case 1, we have

$$A_2 = H_s^r(\mathbb{A}) = [BMO^r(\mathbb{A}), H_q^r(\mathbb{A})]_\theta = [A_1, A_3]_\theta, \quad \frac{\theta}{q} = \frac{1}{s}.$$

Again, applying (5.7), we obtain

$$A_3 = H_q^r(\mathbb{A}) = [A_1, A_4]_\xi = [BMO^r(\mathbb{A}), H_1^r(\mathbb{A})]_{\frac{p}{q}}.$$

The proof of the theorem is complete. ∎

It remains to prove Proposition 5.10. We need the lemma below. Denote

$$\mathcal{Z}_q^r(\mathcal{M}) = \{x \in L_q(\mathcal{M}) : \|x\|_{\mathcal{Z}_q^r} = \sup_{k\geq 1} \|(\mathcal{Z}_{r,k}(x))_{k\geq 1}\|_{L_q(\mathcal{M},\ell_\infty^r)}\} < \infty.$$

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Lemma 5.11. For each $2 < p \le \infty$ and each $\theta \in [0,1]$, the following continuous embedding holds:

$$[\mathcal{Z}^r_\infty(\mathcal{M}), \mathcal{Z}^r_p(\mathcal{M})]_\theta \subset \mathcal{Z}^r_{\frac{p}{\theta}}(\mathcal{M}).$$

Proof. Take $s > 0$ such that $\frac{1}{s} + \frac{1}{p} = \frac{1}{2}$. It follows from Lemma 2.1 that the bilinear map

$$\mathcal{B} : \mathcal{Z}^r_p(\mathcal{M}) \times L_s(\mathcal{M}, \ell^r_2) \to \ell_2(L_2(\mathcal{M}))$$

defined by setting

$$\mathcal{B} : (x, (v_k)_k) \mapsto (\mathcal{Z}_{r,k}(x)^* v_k)_k$$

is a contraction. Particularly,

$$\mathcal{B} : \mathcal{Z}^r_\infty(\mathcal{M}) \times L_2(\mathcal{M}, \ell^r_2) \to \ell_2(L_2(\mathcal{M}))$$

is also a contraction. Then, by the bilinear interpolation result ([5, Theorem 4.4.1]), for each $\theta \in [0,1]$, we have

$$\|T : [\mathcal{Z}^r_\infty(\mathcal{M}), \mathcal{Z}^r_p(\mathcal{M})]_\theta \times [L_2(\mathcal{M}, \ell^r_2), L_s(\mathcal{M}, \ell^r_2)]_\theta \to \ell_2(L_2(\mathcal{M}))\| \le 1.$$

Since $L_\alpha(\mathcal{M}, \ell^r_2)$ is 1-complemented in $L_\alpha(\mathcal{M}\bar{\otimes}B(\ell_2))$ for each $\alpha \ge 1$, it follows that $[L_2(\mathcal{M}, \ell^r_2), L_s(\mathcal{M}, \ell^r_2)]_\theta = L_{s_2}(\mathcal{M}, \ell^r_2)$ with $\frac{1}{s_2} = \frac{1-\theta}{2} + \frac{\theta}{s}$. On the other hand, it is easy to see

$$\frac{1}{s_2} + \frac{\theta}{p} = \frac{1}{2}.$$

Therefore, according to Lemma 2.1, we further obtain

$$[\mathcal{Z}^r_\infty(\mathcal{M}), \mathcal{Z}^r_p(\mathcal{M})]_\theta \subset \mathcal{Z}^r_{\frac{p}{\theta}}(\mathcal{M}).$$

■

Lemma 5.12. If $2 < q \le \infty$, then, for each $x \in L_q(\mathcal{M})$, we have

$$\|(\mathcal{Z}_{r,k}(x))_{k\ge1}\|_{L_q(\mathcal{M},\ell^r_\infty)} \le c_q \|x\|_q.$$

Similarly, we also have

$$\|(\mathcal{Z}_{c,k}(x))_{k\ge1}\|_{L_q(\mathcal{M},\ell^c_\infty)} \le c_q \|x\|_q.$$

Proof. Let $x \in L_q(\mathcal{M})$, and take $p > 0$ satisfying $1/p + 1/q = 1/2$. According to Lemma 2.1, for any given $\epsilon > 0$, we find sequence of $(v_k)_{k\ge1} \subset L_p(\mathcal{M})$ such that

$$\Big\| \Big(\sum_{k\ge1} v_k v_k^* \Big)^{1/2} \Big\|_p \le 1$$

and

$$\|(\mathcal{Z}_{r,k}(x))_{k\ge1}\|_{L_q(\mathcal{M},\ell^r_\infty)} \le \Big(\sum_{k\ge1} \|\mathcal{Z}_{r,k}(x)^* v_k\|_2^2 \Big)^{1/2} + \epsilon.$$

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Observe that

$$\begin{aligned}\sum_{k\geq 1}\|\mathcal{Z}_{r,k}(x)^*v_k\|_2^2 &= \sum_{k\geq 1}\tau(v_k^*e_kx(1-e_{k-1})x^*e_kv_k)\\ &\leq \sum_{k\geq 1}\tau(v_k^*e_k|x^*|^2e_kv_k)\\ &= \tau\Big(|x^*|^2\sum_{k\geq 1}e_kv_kv_k^*e_k\Big) \leq \||x^*|^2\|_{\frac{q}{2}}\Big\|\sum_{k\geq 1}e_kv_kv_k^*e_k\Big\|_{\frac{p}{2}}.\end{aligned}$$

Note that $p\geq 2$. Finally, combining with Corollary 4.5, we obtain

$$\|(\mathcal{Z}_{r,k}(x))_{k\geq 1}\|_{L_q(\mathcal{M},\ell_\infty^r)} \leq c_p\|x\|_q\Big\|\sum_{k\geq 1}v_kv_k^*\Big\|_{\frac{p}{2}}^{1/2}+\epsilon \leq c_q\|x\|_q+\epsilon.$$

Letting $\epsilon\to 0$, the desired assertion follows. ∎

Now we provide the proof of Proposition 5.10.

Proof of Proposition 5.10. Let $2<p<q<\infty$. Using Lemma 5.12 and Lemma 5.11 with $\theta=p/q$, we have

$$L_q(\mathcal{M}) = [L_\infty(\mathcal{M}), L_p(\mathcal{M})]_\theta \subset [\mathcal{Z}_\infty^r(\mathcal{M}), \mathcal{Z}_p^r(\mathcal{M})]_\theta \subset \mathcal{Z}_q^r(\mathcal{M}).$$

According to the definition of $L_p^rMO(\mathbb{A})$ and $\mathcal{Z}_p^r(\mathcal{M})$, this actually implies that

$$H_q^r(\mathbb{A}) \subset [BMO^r(\mathbb{A}), L_p^rMO(\mathbb{A})]_\theta \subset L_q^rMO(\mathbb{A}).$$

On the other hand, by Corollary 5.6, we deduce that

$$H_q^r(\mathbb{A}) = [BMO^r(\mathbb{A}), L_p^rMO(\mathbb{A})]_\theta = L_q^rMO(\mathbb{A}).$$

The proof is complete. ∎

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